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South Pacific albacore: manuscript workflow example

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1 Executive summary

This is a workflow demonstration, not a stock assessment. Diagnostic and stepwise figures use archived fits; other studies include illustrative values. The 2027 cover is an example. Complete the methods, results and interpretation with the current assessment before submission.

Report outputs and interactive supplements are listed in Table 1.

2 Introduction

Assessing South Pacific albacore requires a link between what fisheries observe and how the population changes. Catch records describe removals, abundance indices track relative trends, and size samples help resolve the sizes and ages affected by fishing. Biological studies provide the context needed to interpret these observations.

An integrated assessment brings these sources together in a population model. The analysis can then examine changes in recruitment, spawning potential and fishing mortality, and assess stock status against defined reference points. Differences in data coverage and uncertainty in the underlying processes are central to that interpretation.

Each update revisits the population history as well as recent stock status. New observations and revised assumptions can both change earlier estimates. A stepwise comparison helps explain those changes, while sensitivity analyses show how strongly the conclusions depend on particular choices.

3 Background

3.1 Fisheries

Longline fleets take much of the South Pacific albacore catch, while southern surface fisheries target smaller fish. The fleets include distant-water vessels and vessels based in Pacific Island countries and territories. Troll fishing around New Zealand and the subtropical convergence zone, together with historical driftnet catches, forms part of this broader fishery history (Nikolic et al., 2017; Vidal et al., 2021).

These fisheries sample different parts of the population. Their catches and size distributions reflect where and how they operate, as well as changes in fish abundance. This provides a basis for defining fisheries in the assessment.

3.2 Distribution and movement

Albacore occupy a wide range of tropical and temperate habitats. North and South Pacific populations have generally been considered separate stocks, while structure within the South Pacific remains less certain (Anderson et al., 2019; Nikolic et al., 2017). Regional differences in growth, genetics and otolith chemistry help describe that structure (Anderson et al., 2019; Macdonald et al., 2013; Williams et al., 2012).

Tag recoveries and patterns in fish size suggest movement between southern feeding grounds and warmer waters as fish grow. Smaller albacore are common in cooler waters, whereas mature fish occur farther north during the spawning season (Farley et al., 2014; Nikolic et al., 2017). Tagging evidence is uneven across the region because releases, fishing effort and reporting vary in space and time. Seasonal catch patterns provide additional information on distribution (Langley, 2004).

Habitat-based models offer hypotheses about movement and the distribution of recruitment (Lehodey et al., 2015; Senina et al., 2020). These can inform assessment alternatives alongside evidence from fisheries and biological sampling. Region boundaries provide a way to represent spatial variation; they do not necessarily delimit closed populations.

3.3 Depth use and availability to fishing

The depths occupied by albacore affect their availability to different gears. Longline catches tend to come from deeper water in warmer areas, while troll fisheries sample fish near the surface (Bigelow et al., 2006). Electronic tagging has also shown changes in depth through the day and across latitudes (Williams et al., 2015). These behaviours help explain why catchability can vary even when abundance does not.

3.4 Growth

Otolith ages and length-frequency patterns provide complementary views of growth. Surface fisheries sample younger fish, and growth slows as albacore become larger (Farley et al., 2013a;

Williams et al., 2012). Differences between sexes and across the South Pacific add variation around the mean growth curve.

The interpretation of age data also depends on ageing methods and the timing of sampling within the year (Farley et al., 2021). Comparisons of growth estimates therefore need to account for which fish were sampled and how their ages were assigned. This is particularly relevant when combining age data with modes in length-frequency samples.

3.5 Reproduction

South Pacific albacore spawn in warm waters during the austral summer. Maturity and reproductive activity vary with latitude and season (Farley et al., 2013b; Farley et al., 2014). A reproductive ogive summarises how the contribution to spawning changes with fish size or age.

Converting that ogive to population spawning potential depends on the treatment of maturity, female proportion, body mass and, where used, fecundity. The resulting quantity may be expressed as biomass or as a relative measure of reproductive output. Its definition is given in Section 5.2.4.

3.6 Natural mortality

Natural mortality is difficult to estimate directly from fishery data. Life history and relationships between mortality and body size can help define plausible age-dependent patterns (Lorenzen, 1996). When those patterns depend on growth, the two sets of assumptions need to be considered together. Their effects on population scale and stock status can then be examined through sensitivity analyses.

4 Data compilation

4.1 Assessment area and data period

Spatial structure links the population being assessed to the areas sampled by fisheries. Population regions describe where fish live and move; fishery strata group observations from similar fishing operations. An areas-as-fleets approach can retain differences among fishing areas within a pooled population (Waterhouse et al., 2014).

The assessment area and model regions are shown in Figure 1. Coverage by data type, fishery and year is summarised in Figure 5.

4.2 Fisheries and catch

Fishery definitions group observations with similar selectivity and fishing practices. Gear, area, fleet and changes in operations can all inform these groups. Extraction fisheries account for removals, while abundance-index series describe relative population trends.

Fishery definitions and observation units are listed in Table 2. Annual catches are summarised by gear in Figure 2. Regional catch histories are shown in Figure 3. The spatial distribution of catch is shown in Figure 4.

4.3 Abundance indices

Catch per unit effort (CPUE) reflects both fish abundance and the way a fleet operates. Standardisation aims to account for changes in fishing practices and sampling so that the remaining trend can inform relative abundance. Spatiotemporal methods also account for spatial dependence and combine predictions over a defined area (Thorson et al., 2015). Combining fleets has helped extend the coverage of earlier albacore analyses (Vidal et al., 2021).

Spatial patterns in nominal CPUE are shown in Figure 6.

4.4 Length composition

Length samples describe the size distribution of fish in the catch. Observer and port-sampling programmes provide much of this information, but their coverage varies among fleets and periods. Fish measured in the same sampling event may also be more alike than fish drawn independently from the catch. Sample construction and likelihood weighting therefore need to be considered together.

4.5 Age observations

Age-length data link measured fish sizes to population age structure. A length-at-age model describes $P(L \mid A)$; an age-at-length model describes $P(A \mid L)$. The second also depends on the age distribution of the sampled fish. Bayes' rule expresses the connection:

$$P(A = a \mid L = l) = \frac{P(L = l \mid A = a)\, \pi_a}{\sum_b P(L = l \mid A = b)\, \pi_b},$$

where π_a represents the sampled age distribution, including selectivity where appropriate.

5 Model description

5.1 Model framework

The model connects population change to fishery removals and the observations used in fitting. Population processes determine the numbers of fish available; fishing processes determine which fish are caught; observation models describe how the data relate to those quantities. MULTIFAN-CL is an established example of this integrated approach (Fournier et al., 1998; Hampton and Fournier, 2001; Kleiber et al., 2019).

5.2 Population processes

5.2.1 Recruitment and initial population

Recruitment adds fish to the youngest modelled age class. Its definition links that age class to the timing and spatial distribution of entry into the population. Variation in recruitment can be represented around a mean or a stock-recruitment relationship.

For a Beverton-Holt relationship, steepness describes recruitment at 20% of unfished equilibrium spawning potential relative to unfished equilibrium recruitment (Francis, 1992). This parameter influences the stock's assumed capacity to replenish itself as spawning potential declines.

5.2.2 Growth and natural mortality

Growth translates population age structure into a distribution of fish lengths. A weight-length relationship then connects fish size to biomass. Variation around mean length at age is important because fish of the same age can occur in several length classes.

Natural mortality governs survival in the absence of fishing. Its time unit must be consistent with the population model, and any dependence on growth must be carried through the biological inputs.

Growth inputs and estimates are shown in Figure 7. Natural mortality at age is shown in Figure 9.

5.2.3 Movement

In a spatial model, movement redistributes fish among regions. The resulting transitions depend on age, season and the sequence of population processes within a time step. Movement rates and transition probabilities describe different parts of that calculation.

Regional movement probabilities are shown in Figure 33.

5.2.4 Reproductive potential

Spawning potential combines the population at each age or size with its contribution to reproduction. The reproductive ogive determines that contribution and the units of the resulting population measure.

The reproductive ogive is shown in Figure 8.

5.3 Fishing processes

Selectivity describes the relative susceptibility of fish of different sizes or ages to a fishery. Catchability links an index to the population it samples or, in an effort-based formulation, links effort to fishing mortality. Their parameterisation determines how fishing operations enter the population model.

Fishery selectivity at age is shown in Figure 31. Selectivity expressed by length is shown in Figure 32.

5.4 Observation models

Each observation model connects a data source to its predicted quantity and specifies how deviations contribute to the objective. For composition data, the information assigned to a sample depends on its effective sample size and the chosen likelihood. These choices affect the balance among data sources.

5.5 Parameter estimation

Fitting minimises the negative log-likelihood and any additional penalties. Automatic differentiation supplies derivatives for optimisation (Fournier et al., 2012). Numerical convergence and the information available to estimate parameters are examined in Section 7.

6 Model development

6.1 Stepwise development

A stepwise sequence separates the effects of successive changes in data and model assumptions. Comparing both population estimates and data fits helps explain why the updated assessment differs from its starting model.

The sequence of population estimates is shown in Figure 10.

6.2 Sensitivity analyses

Sensitivity analyses examine assumptions that could affect the assessment's conclusions. Changing one assumption at a time clarifies its influence; combinations of changes reveal interactions. The alternatives should reflect the uncertainties relevant to the data and model.

Sensitivity to stock-recruitment steepness is shown in Figure 41. Sensitivity to natural mortality is shown in Figure 42. The effect of regional index scaling is shown in Figure 43.

6.3 Model ensemble

An ensemble brings the selected assumptions together. Its distribution of results depends on the models retained, their weights and the treatment of uncertainty within each model. These decisions define what the ensemble summaries represent.

The ensemble design and model retention are summarised in Figure 44.

7 Model diagnostics

7.1 Starting-value trials (jitter)

Jitter trials refit the same model and data from different starting values. Agreement in final objectives and population estimates indicates that the solution is reproducible across those trials. Gradients help distinguish converged solutions from runs that stopped before reaching an optimum.

Objectives and gradients are compared in Figure 19. The corresponding population estimates are shown in Figure 20. Regional depletion is compared in Figure 21.

7.2 Simulation-estimation tests (self-test)

A self-test generates data from a known model and then estimates that model from the simulated observations. Comparing the estimates with their generating values reveals bias and variability in recovery under the assumed processes. Where uncertainty intervals are available, their coverage can also be examined.

Recovery of selected quantities is summarised in Figure 22. Recovery through time is shown in Figure 23.

7.3 Retrospective analysis

Retrospective analysis examines how estimates change as terminal observations are removed. Each reduced-data fit is compared with the full-data fit over their shared years (Cadigan and Farrell, 2005). Repeated shifts in the same direction can reveal a systematic pattern in recent estimates.

Population trajectories for the retrospective fits are compared in Figure 24.

7.4 Hindcast evaluation

Hindcasts assess predictions against observations withheld from fitting. At each cutoff, the model is fitted using the information available up to that point and predicts subsequent observations. The comparison measures predictive performance at the specified horizons.

Predictions and withheld observations are compared in Figure 54. Prediction errors and skill measures are summarised in Table 7.

7.5 Likelihood profiles and identifiability

A profile follows the objective as one parameter or derived quantity is fixed at a sequence of values and the remaining parameters are re-estimated. Separating the objective by data source shows where information is concentrated and where sources favour different values. Total average biomass is one possible measure of population scale for this analysis (McKechnie et al., 2017; Tremblay-Boyer et al., 2017).

The total profile and contributions by data source are shown in Figure 25. Index contributions are shown in Figure 26. Length-composition contributions are shown in Figure 27. Age-data contributions are shown in Figure 28.

8 Results

8.1 Data fit

Abundance-index fits and residuals are shown in Figure 11. Pooled length-composition fits are compared by fishery in Figure 12. Regional length-composition fits are shown in Figure 15.

Length summaries through time for longline fisheries are shown in Figure 16. Length summaries for other fisheries are shown in Figure 17. Conditional age-at-length fits are shown in Figure 18.

8.2 Population estimates

The diagnostic model's population trajectories are shown in Figure 30. Annual recruitment is shown by region and overall in Figure 35. Recruitment allocation among regions and seasons is shown in Figure 36. The relationship between spawning potential and recruitment is shown in Figure 37. Regional contributions to population quantities are shown in Figure 38. Biomass under fitted fishing and the no-fishing calculation is compared in Figure 39. Juvenile and adult fishing mortality is shown in Figure 40.

8.3 Robustness and uncertainty

The comparison with an age-structured production model is shown in Figure 29. Trajectories from the retained ensemble models are shown in Figure 46. Ensemble estimates and uncertainty are summarised in Figure 47. Stock-status variation across biological assumptions is shown in Figure 45.

9 Stock status and interpretation

9.1 Depletion and fishery impact

Dynamic depletion measures spawning potential under fishing relative to a modelled population without fishing at the same time:

$$D_t = \frac{SB_t}{SB_{t,F=0}},$$

where SB_t is spawning potential under the fitted fishing history and $SB_{t,F=0}$ is its no-fishing counterpart. The comparison accounts for the population variation represented in both trajectories. Its interpretation depends on how recruitment is treated in the no-fishing calculation.

Management measures may use averages over specified periods instead. The numerator, denominator and time windows then define the quantity being reported. Averaging depletion ratios and taking a ratio of mean spawning potentials generally give different results.

Fishery impacts on spawning potential are shown in Figure 51.

9.2 Yield and reference points

Equilibrium yield analysis compares the long-term catch supported by different levels of fishing under a stated fishing pattern and stock-recruitment relationship. Its maximum defines MSY, with associated quantities F_{MSY} and SB_{MSY} (Kleiber et al., 2019). These reference points depend on the biological assumptions and the ages exposed to fishing.

Equilibrium yield is shown in Figure 52. Changes in yield reference points under historical fishing patterns are shown in Figure 50. Reference-point estimates and definitions are given in Table 5.

9.3 Stock status

Stock-status summaries relate estimated spawning potential and fishing mortality to the chosen reference points. Kobe plots use SB/SB_{MSY} and F/F_{MSY}; Majuro plots use a defined depletion measure alongside F/F_{MSY}. Ensemble probabilities depend on the weighting of models and the uncertainty included in the comparison.

Management quantities are summarised in Table 3. Probabilities relative to the reference points are given in Table 4. The stock-status comparison is shown in Figure 48. Changes in stock status through time are shown in Figure 49.

10 Projections

Projections examine how the population could change under specified fishing and recruitment scenarios. The outcomes depend on the control applied, such as catch or fishing mortality, and on the sources of uncertainty carried forward from the assessment.

10.1 Scenarios

10.2 Projected outcomes

Historical and projected trajectories are shown in Figure 53. Projected management quantities are summarised in Table 6.

11 Discussion

11.1 Interpretation

11.2 Limitations

11.3 Conclusions and research priorities

12 Acknowledgements

13 References

Anderson, G., Hampton, J., Smith, N., and Rico, C. (2019). Indications of strong adaptive population genetic structure in albacore tuna (*Thunnus alalunga*) in the southwest and central Pacific Ocean. *Ecol Evol*, 9(18): 10354–10364 https://doi.org/10.1002/ece3.5554.

Bigelow, K., Musyl, M. K., Poisson, F., and Kleiber, P. (2006). Pelagic longline gear depth and shoaling. *Fisheries Research*, 77(2): 173–183 https://doi.org/10.1016/j.fishres.2005.10.010.

Cadigan, N. G., and Farrell, P. J. (2005). Local influence diagnostics for the retrospective problem in sequential population analysis. *ICES Journal of Marine Science: Journal du Conseil*, 62(2):

256–265 https://doi.org/10.1016/j.icesjms.2004.11.015.

Farley, J. H., Hoyle, S. D., Eveson, J. P., Williams, A. J., Davies, C. R., and Nicol, S. J. (2014). Maturity Ogives for South Pacific Albacore Tuna (Thunnus alalunga) That Account for Spatial and Seasonal Variation in the Distributions of Mature and Immature Fish. *PLoS ONE*, 9(1): e83017 https://doi.org/10.1371/journal.pone.0083017.

Farley, J., Krusic-Golub, K., and Eveson, P. (2021). Ageing of South Pacific Albacore: (Project 106). WCPFC-SC17-SA-IP-10.

Farley, J. H., Williams, A. J., Clear, N. P., Davies, C. R., and Nicol, S. J. (2013a). Age estimation and validation for South Pacific albacore (Thunnus alalunga). *Journal of Fish Biology*, 82(5): 1523–1544 https://doi.org/10.1111/jfb.12077.

Farley, J. H., Williams, A. J., Hoyle, S. D., Davies, C. R., and Nicol, S. J. (2013b). Reproductive Dynamics and Potential Annual Fecundity of South Pacific Albacore Tuna (Thunnus alalunga). *PLoS ONE*, 8(4): e60577.

Fournier, D. A., Hampton, J., and Sibert, J. R. (1998). MULTIFAN-CL: A length-based, age-structured model for fisheries stock assessment, with application to South Pacific albacore, Thunnus alalunga. *Canadian Journal of Fisheries and Aquatic Sciences*, 55: 2105–2116.

Fournier, D. A., Skaug, H. J., Ancheta, J., Ianelli, J., Magnusson, A., Maunder, M. N., Nielson, A., and Sibert, J. (2012). AD Model Builder: Using automatic differentiation for statistical inference of highly parameterized complex nonlinear models. *Optimization Methods and Software*, 27(2): 233–249.

Francis, R. I. C. C. (1992). Use of risk analysis to assess fishery management strategies: A case study using orange roughy (Hoplostethus atlanticus) on the Chatham Rise, New Zealand. *Canadian Journal of Fisheries and Aquatic Sciences*, 49: 922–930 https://doi.org/10.1139/f92-102.

Hampton, J., and Fournier, D. A. (2001). A spatially-disaggregated, length-based, age-structured population model of yellowfin tuna (Thunnus albacares) in the western and central Pacific Ocean. *Marine and Freshwater Research*, 52: 937–963.

Kleiber, P., Fournier, D., Hampton, J., Davies, N., Bouye, F., and Hoyle, S. (2019). MULTIFAN-CL User's Guide.

Langley, A. (2004). An examination of the influence of recent oceanographic conditions on the catch rate of albacore in the main domestic longline fisheries. Working Paper SA-4., 17th Standing Committee on Tuna; Billfish. 9–18 August 2004. Majuro, Republic of Marshall Islands.

Lehodey, P., Senina, I., Nicol, S., and Hampton, J. (2015). Modelling the impact of climate change on South Pacific albacore tuna. *Deep Sea Research Part II: Topical Studies in Oceanography*, 113: 246–259 https://doi.org/10.1016/j.dsr2.2014.10.028.

Lorenzen, K. (1996). The relationship between body weight and natural mortality in juvenile and adult fish: A comparison of natural ecosystems and aquaculture. *Journal of Fish Biology*, 49(4): 627–642 https://doi.org/10.1111/j.1095-8649.1996.tb00060.x.

Macdonald, J. I., Farley, J. H., Clear, N. P., Williams, A. J., Carter, T. I., Davies, C. R., and Nicol, S. J. (2013). Insights into mixing and movement of South Pacific albacore Thunnus alalunga derived from trace elements in otoliths. *Fisheries Research*, 148: 56–63 https://doi.org/10.1016/j.fishres.2013.08.004.

McKechnie, S., Pilling, G., and Hampton, J. (2017). Stock assessment of bigeye tuna in the western and central Pacific Ocean. WCPFC-SC13-2017/SA-WP-05, Rarotonga, Cook Islands, 9–17 August 2017.

Nikolic, N., Morandeau, G., Hoarau, L., West, W., Arrizabalaga, H., Hoyle, S., Nicol, S. J., Bourjea, J., Puech, A., Farley, J. H., Williams, A. J., and Fonteneau, A. (2017). Review of albacore tuna, Thunnus alalunga, biology, fisheries and management. *Rev Fish Biol Fisheries*, 27(4): 775–810 https://doi.org/10.1007/s11160-016-9453-y.

Senina, I. N., Lehodey, P., Hampton, J., and Sibert, J. (2020). Quantitative modelling of the spatial dynamics of South Pacific and Atlantic albacore tuna populations. *Deep Sea Research Part II: Topical Studies in Oceanography*, 175: 104667 https://doi.org/10.1016/j.dsr2.2019.104667.

Thorson, J. T., Shelton, A. O., Ward, E. J., and Skaug, H. J. (2015). Geostatistical delta-generalized linear mixed models improve precision for estimated abundance indices for West Coast groundfishes. *Ices Journal of Marine Science*, 72(5): 1297–1310 https://doi.org/10.1093/icesjms/fsu243.

Tremblay-Boyer, L., McKechnie, S., Pilling, G., and Hampton, J. (2017). Stock assessment of yellowfin tuna in the Western and Central Pacific Ocean. WCPFC-SC13-2017/SA-WP-06, Rarotonga, Cook Islands, 9-17 August 2017.

Vidal, T., Castillo Jordan, C., Peatman, T., Ducharme-Barth, N., Xu, H., Williams, P., Lennert-Cody, C., and Hamer, P. (2021). Background analysis and data inputs for the 2021 South Pacific albacore tuna stock assessment. WCPFC-SC17-SA-IP-03.

Waterhouse, L., Sampson, D. B., Maunder, M., and Semmens, B. X. (2014). Using areas-as-fleets selectivity to model spatial fishing: Asymptotic curves are unlikely under equilibrium conditions. *Fisheries Research*, 158: 15–25 https://doi.org/10.1016/j.fishres.2014.01.009.

Williams, A. J., Allain, V., Nicol, S. J., Evans, K. J., Hoyle, S. D., Dupoux, C., Vourey, E., and Dubosc, J. (2015). Vertical behavior and diet of albacore tuna (Thunnus alalunga) vary with latitude in the South Pacific Ocean. *Deep Sea Research Part II: Topical Studies in Oceanography*, 113: 154–169 https://doi.org/10.1016/j.dsr2.2014.03.010.

Williams, A. J., Farley, J. H., Hoyle, S. D., Davies, C. R., and Nicol, S. J. (2012). Spatial and sex-specific variation in growth of albacore tuna (Thunnus alalunga) across the South Pacific Ocean. *PLoS ONE*, 7(6): e39318.

14 Tables

Table 1: Public report, interactive supplements and published outputs for this workflow demonstration.

Resource	Figures and tables	Interactive output
Workflow demonstration	PDF report	HTML report
Diagnostic fit	HTML supplement	Model viewer
Stepwise development	HTML supplement	Compare development steps

Resource	Figures and tables	Interactive output
Jitter	HTML supplement	Model viewer
Self-test	HTML supplement	Model viewer
Retrospective analysis	HTML supplement	Model viewer
Sensitivities	HTML supplement	Model viewer
Uncertainty ensemble	HTML supplement	Model viewer
Objective profiles	HTML supplement	Model viewer
Additional output examples	HTML supplement	Model viewer
Hindcasting	HTML supplement	Model viewer
Projection scenarios	HTML supplement	Model viewer
Published outputs	Public repository	Rendered artifacts

Table 2: Fishery definitions and observation units.

Fishery	Region	Role	First year	Last year	Unit	Fishery name	Gear long	Method	Code	Gear	Flag	Area
1	1	Catch	1954	2022	fish	1.LL.DWFN.1a	LL	L	1.LL.DWFN.1a	LL	DWFN	1
2	1	Catch	1954	2022	fish	2.LL.DWFN.1b	LL	L	2.LL.DWFN.1b	LL	DWFN	1
3	1	Catch	1954	2022	fish	3.LL.DWFN.1c	LL	L	3.LL.DWFN.1c	LL	DWFN	1
4	1	Catch	1954	2022	fish	4.LL.DWFN.1d	LL	L	4.LL.DWFN.1d	LL	DWFN	1
5	1	Catch	1992	2022	fish	5.LL.PICT.1ab	LL	L	5.LL.PICT.1ab	LL	PICT	1
6	1	Catch	1982	2022	fish	6.LL.PICT.1ab	LL	L	6.LL.PICT.1ab	LL	PICT	1
7	1	Catch	1987	2022	fish	7.LL.AZ.1ab	LL	L	7.LL.AZ.1ab	LL	AU/NZ	1
8	1	Catch	1954	2022	fish	8.LL.DWFN.1cd	LL	L	8.LL.DWFN.1cd	LL	DWFN	1
9	1	Catch	1984	2022	fish	9.LL.PICT.1cd	LL	L	9.LL.PICT.1cd	LL	PICT	1
10	1	Catch	1985	2022	fish	10.LL.AZ.1cd	LL	L	10.LL.AZ.1cd	LL	AU/NZ	1
11	1	Catch	1982	2022	t	11.TR.ALL.1c	TR	L	11.TR.ALL.1c	TR	ALL	1
12	1	Catch	1987	2022	t	12.TR.ALL.1d	TR	L	12.TR.ALL.1d	TR	ALL	1
13	1	Catch	1983	1990	t	13.DN.ALL.1cd	DN	L	13.DN.ALL.1cd	DN	ALL	1
14	2	Catch	1957	2022	fish	14.LL.EPO.2a	LL	L	14.LL.EPO.2a	LL	ALL	2
15	2	Catch	1961	2021	fish	15.LL.EPO.2b	LL	L	15.LL.EPO.2b	LL	ALL	2
16	2	Catch	1963	2021	fish	16.LL.EPO.2c	LL	L	16.LL.EPO.2c	LL	ALL	2
17	2	Catch	1986	2006	t	17.TR.EPO.2abc	TR	L	17.TR.EPO.2abc	TR	ALL	2
18	1	Index	1954	2022	relative index	18.L.INDEX.1abcd	L	L	18.L.INDEX.1abcd	L	INDEX	1
19	1	Index	1992	2022	relative index	19.TR.INDEX.1ef	TR	L	19.TR.INDEX.1ef	TR	INDEX	1

Fishery	Region	Role	First year	Last year	Unit	Fishery name	Gear long	Method	Code	Gear	Flag	Area
20	2	Index	1954	2022	relative index	20.L.INDEX.2abc	L	L	20.L.INDEX.2abc	L	INDEX	2

Table 3: Management quantities and supplied uncertainty summaries.

Quantity	Period	Median	50% interval	80% interval	95% interval
$SB/SB_{F=0}$	2016–2020	0.45	0.41–0.49	0.37–0.53	0.33–0.57
F/F_{MSY}	2016–2020	0.68	0.61–0.75	0.54–0.82	0.46–0.90
SB/SB_{MSY}	2016–2020	1.35	1.22–1.48	1.08–1.62	0.94–1.76

Table 4: Stock-status probabilities for the declared reference points.

Condition	Probability
$SB/SB_{F=0} < 0.20$	0.01
$F/F_{MSY} > 1$	0.05
$SB/SB_{MSY} < 1$	0.04

Table 5: Model reference points and their definitions.

Quantity	Median	Lower	Upper	Unit
MSY	60000	48000	72000	t/year
F_{MSY}	0.3	0.24	0.36	per year
SB_{MSY}	1000000	800000	1200000	t
SB_{MSY}/SB_0	0.28	0.22	0.34	ratio

Table 6: Illustrative projected quantities in the final projection year; medians and central 80% intervals.

Scenario	Quantity	Year	Median	Lower	Upper	Unit
fixed-catch	biomass depletion	2052	0.9499	0.9473	0.952	ratio
fixed-catch	catch	2052	45150	45150	45150	t
fixed-catch	fishing mortality	2052	0.01518	0.01514	0.01522	per year
fixed-catch	total biomass	2052	2997000	2989000	3004000	t

Scenario	Quantity	Year	Median	Lower	Upper	Unit
higher-F	biomass depletion	2052	0.7427	0.7378	0.7519	ratio
higher-F	catch	2052	180200	179000	182400	t
higher-F	fishing mortality	2052	0.08	0.08	0.08	per year
higher-F	total biomass	2052	2343000	2328000	2372000	t
lower-F	biomass depletion	2052	0.8678	0.8609	0.8719	ratio
lower-F	catch	2052	107400	106500	107900	t
lower-F	fishing mortality	2052	0.04	0.04	0.04	per year
lower-F	total biomass	2052	2738000	2716000	2751000	t

Table 7: Held-out prediction bias (predicted minus observed), RMSE and MASE by series and forecast lead. N counts scored forecasts; MASE uses each cutoff's training naive-error scale.

Series	Lead	N	Bias	RMSE	MASE	Unit
Index 18 (Region 1)	1	5	-0.08847	0.10778	0.39718	relative
Index 18 (Region 1)	2	4	-0.095363	0.117	0.42491	relative
Index 18 (Region 1)	3	3	-0.093805	0.12075	0.41723	relative
Index 19 (Region 1)	1	5	-0.0691	0.18227	0.52469	relative
Index 19 (Region 1)	2	4	-0.0245	0.13014	0.41413	relative
Index 19 (Region 1)	3	3	-0.026698	0.1508	0.49854	relative
Index 20 (Region 2)	1	5	-0.2195	0.25108	0.84494	relative
Index 20 (Region 2)	2	4	-0.26202	0.28574	0.99863	relative
Index 20 (Region 2)	3	3	-0.30353	0.32467	1.1481	relative

15 Figures

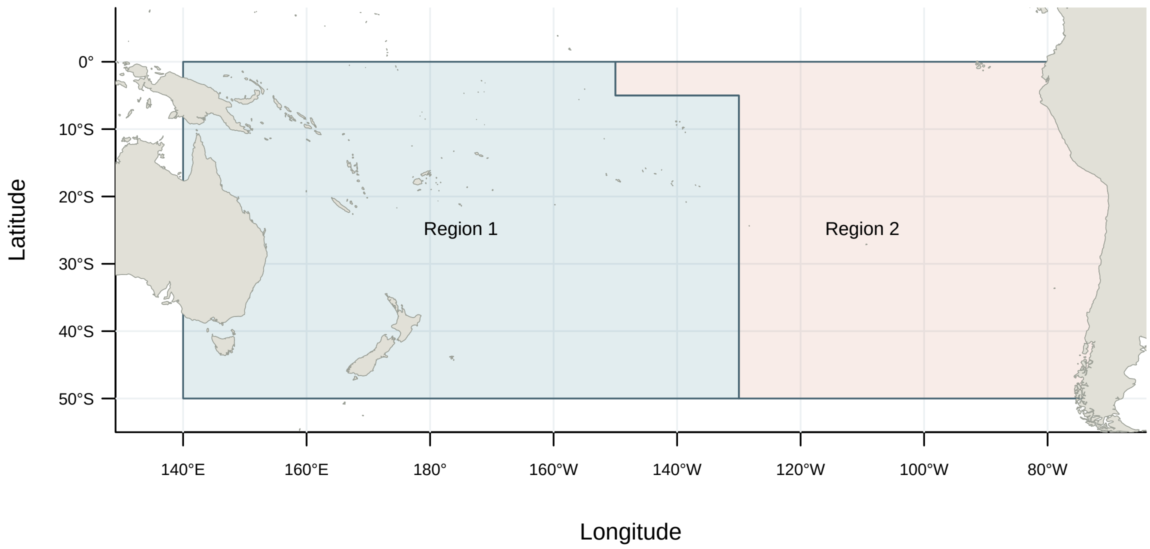

Figure 1: Model regions for the 2024 South Pacific albacore assessment (WCPFC-SC20-SA-WP-02, Rev.03, Figure 2).

Figure 2: Illustrative output. Catch by fishing group.

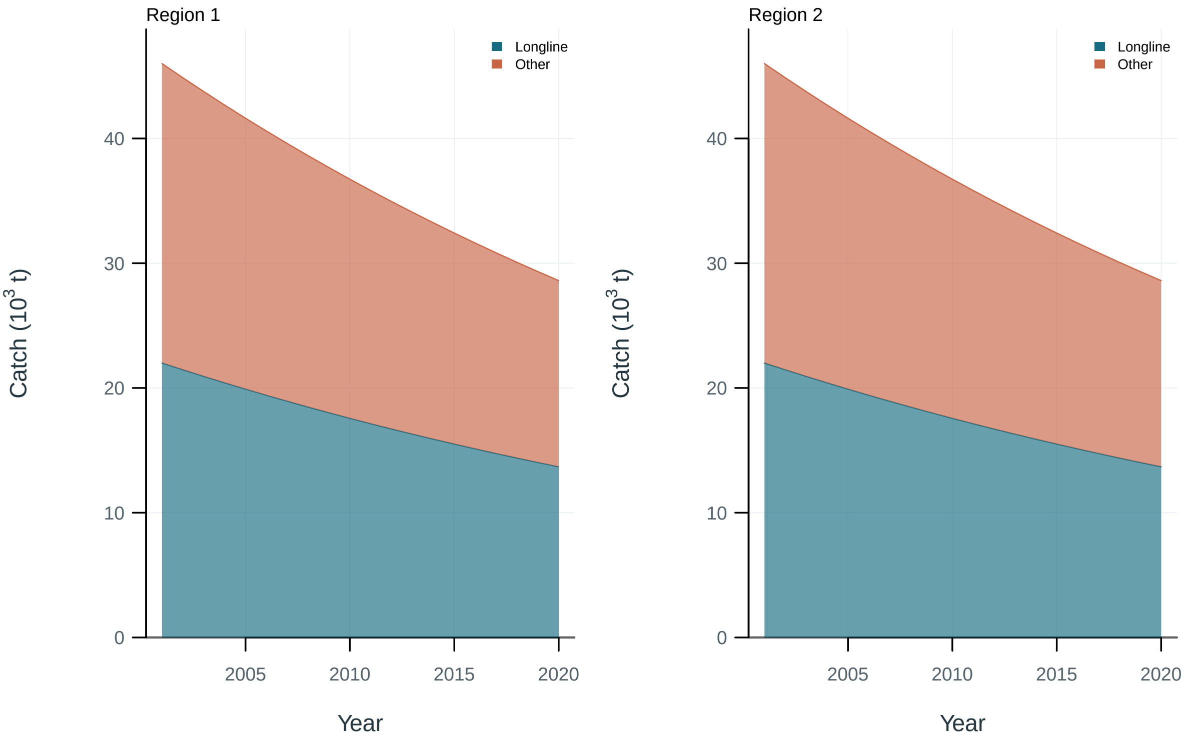

Figure 3: Illustrative output. Catch by fishing group and region.

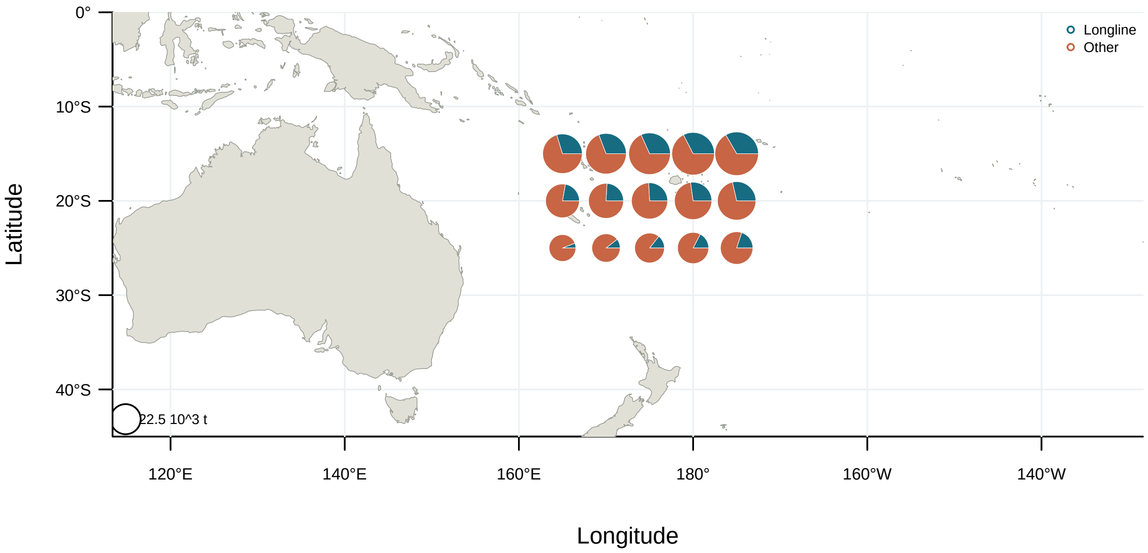

Figure 4: Illustrative output. Spatial catch distribution by fishing group.

Figure 5: Observation availability by data type and fishery. Coloured cells indicate years with observations.

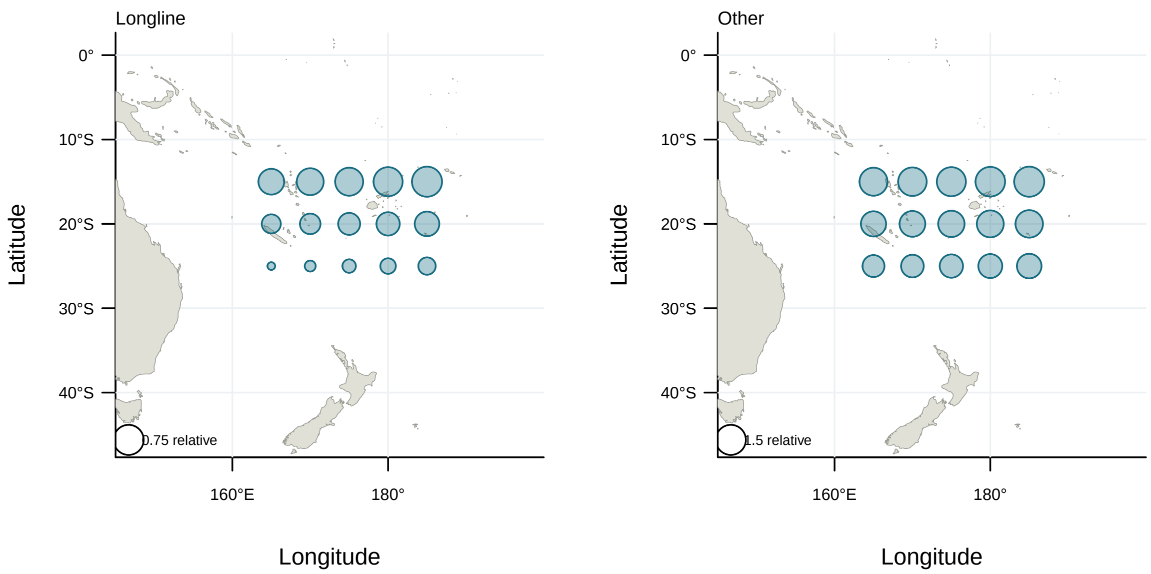

Figure 6: Illustrative output. Spatial distribution of nominal abundance indices.

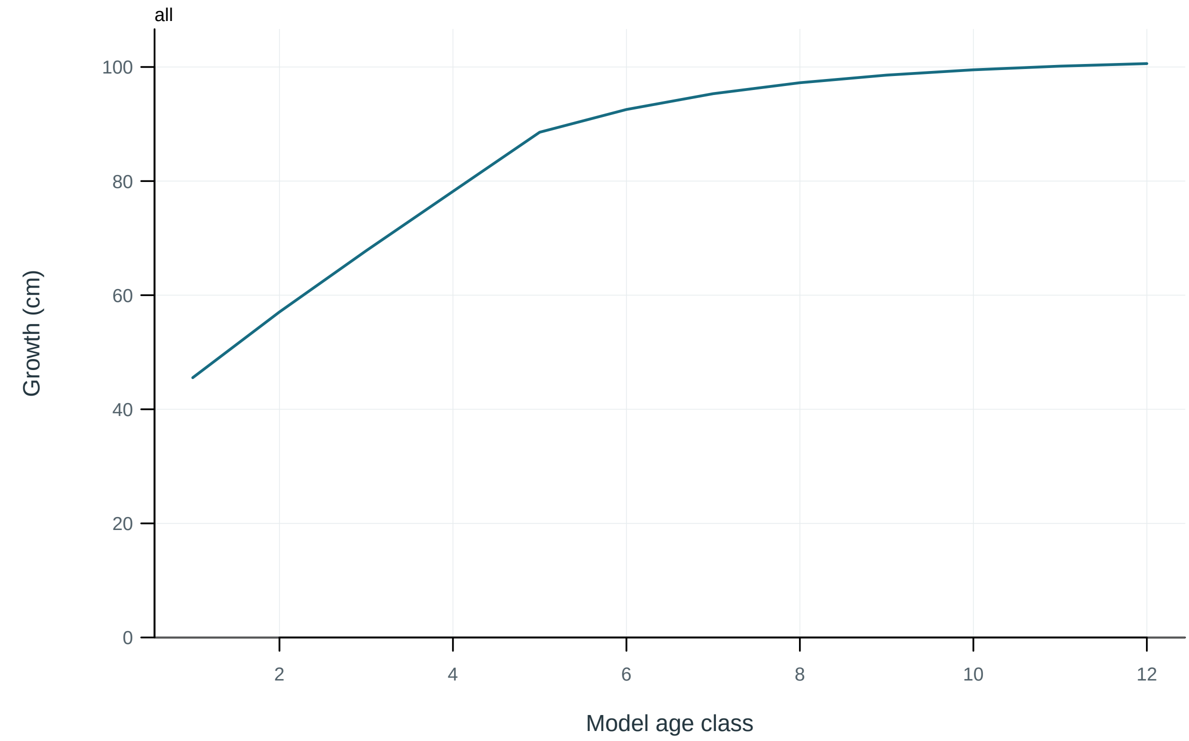

Figure 7: Mean length at the start of each model age class.

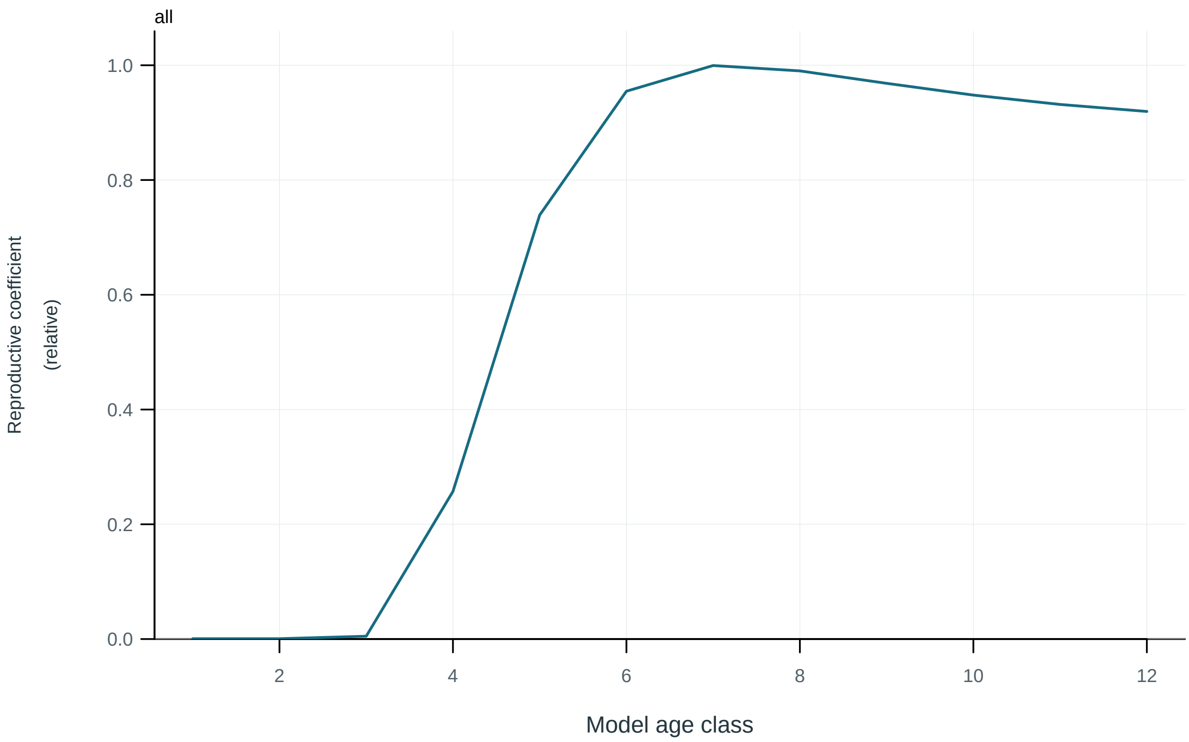

Figure 8: Model reproductive coefficient at age.

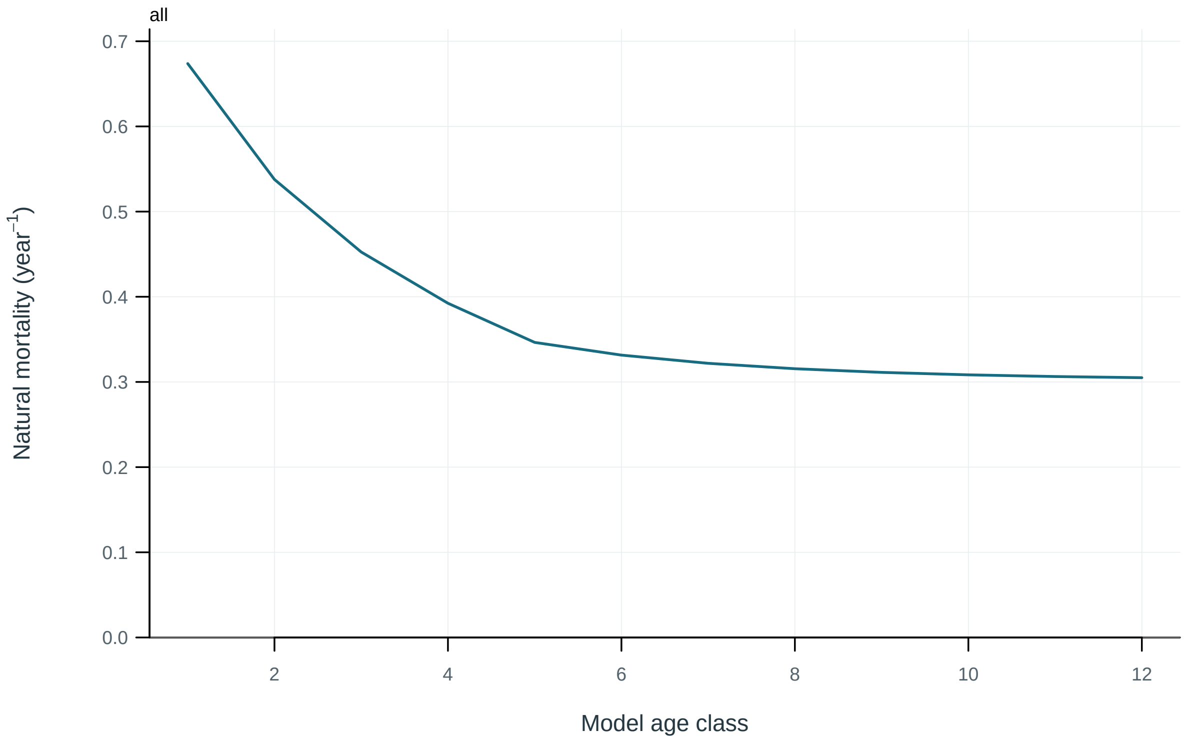

Figure 9: Annual instantaneous natural mortality at age.

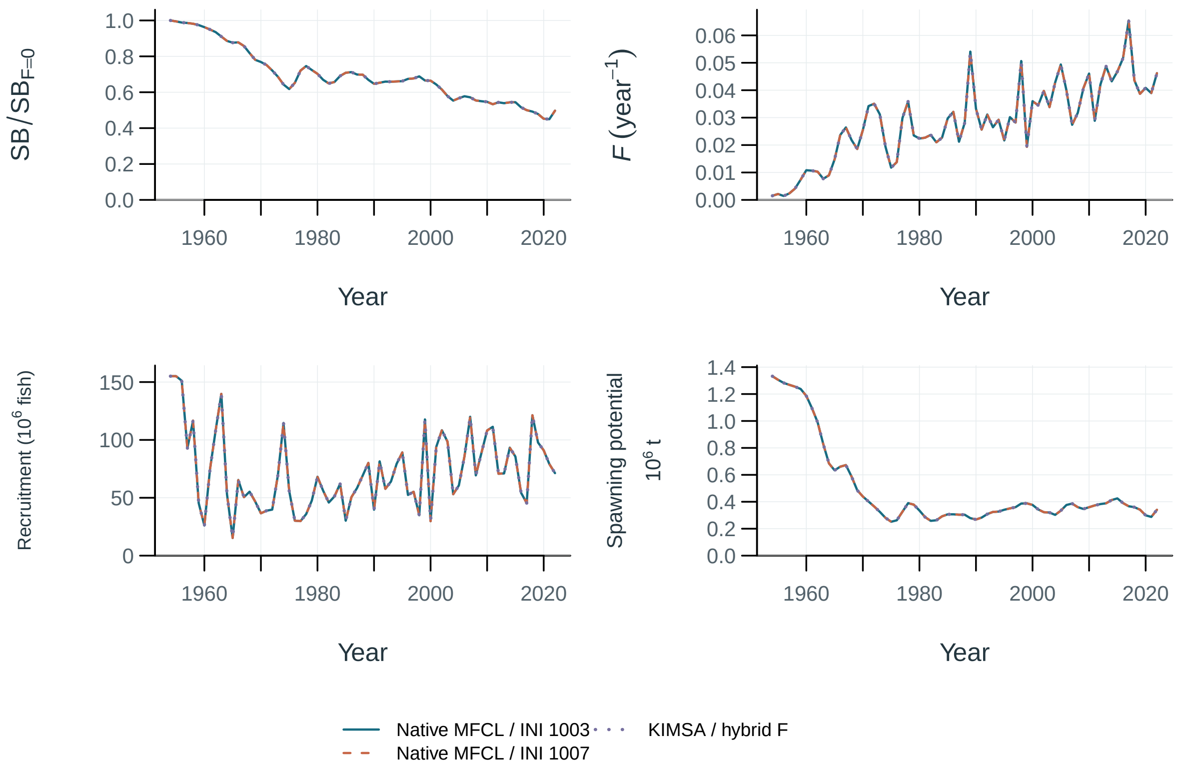

Figure 10: Stepwise model trajectories.

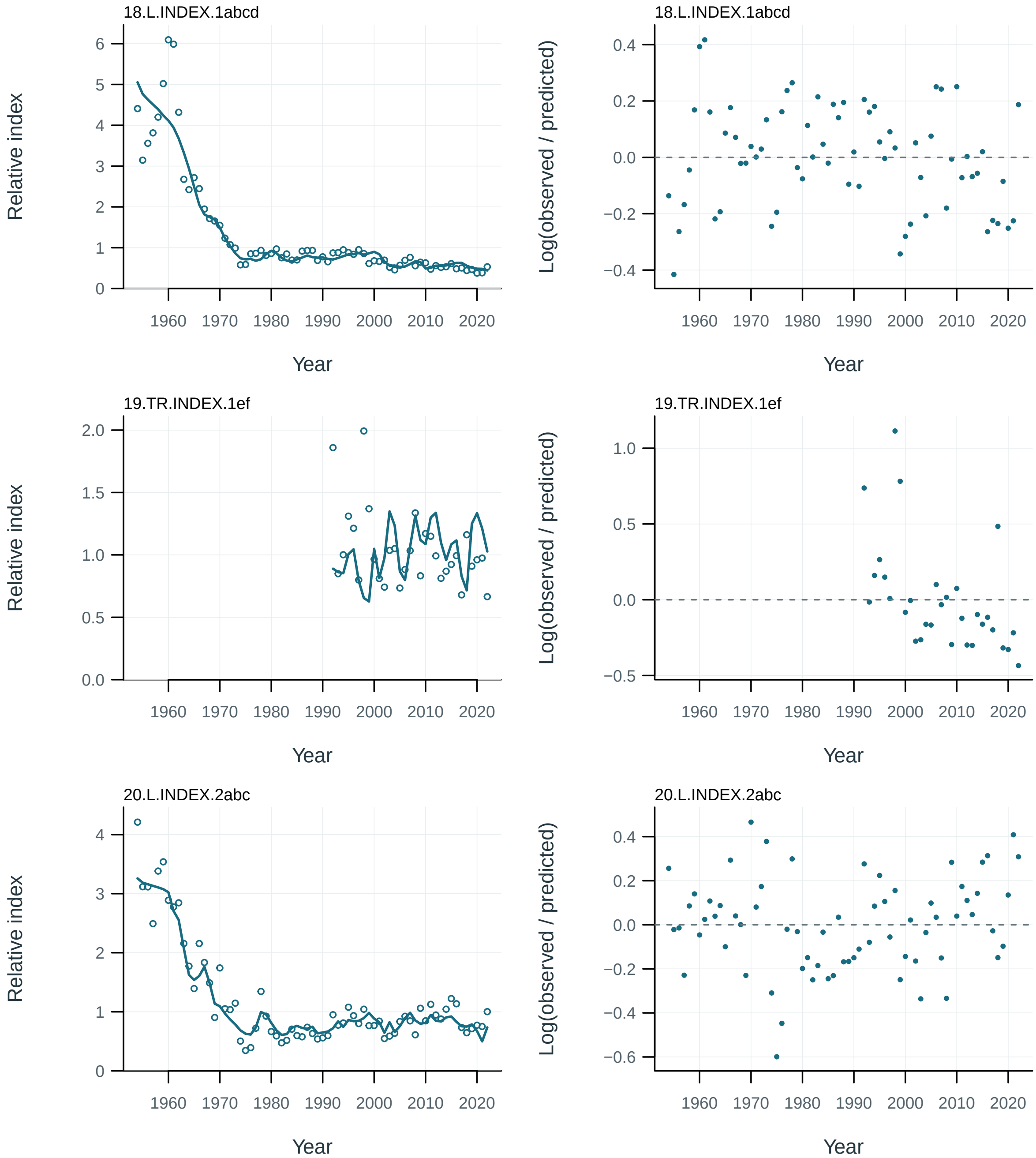

Figure 11: Abundance-index fits and log residuals by series. Residuals are log(observed/predicted).

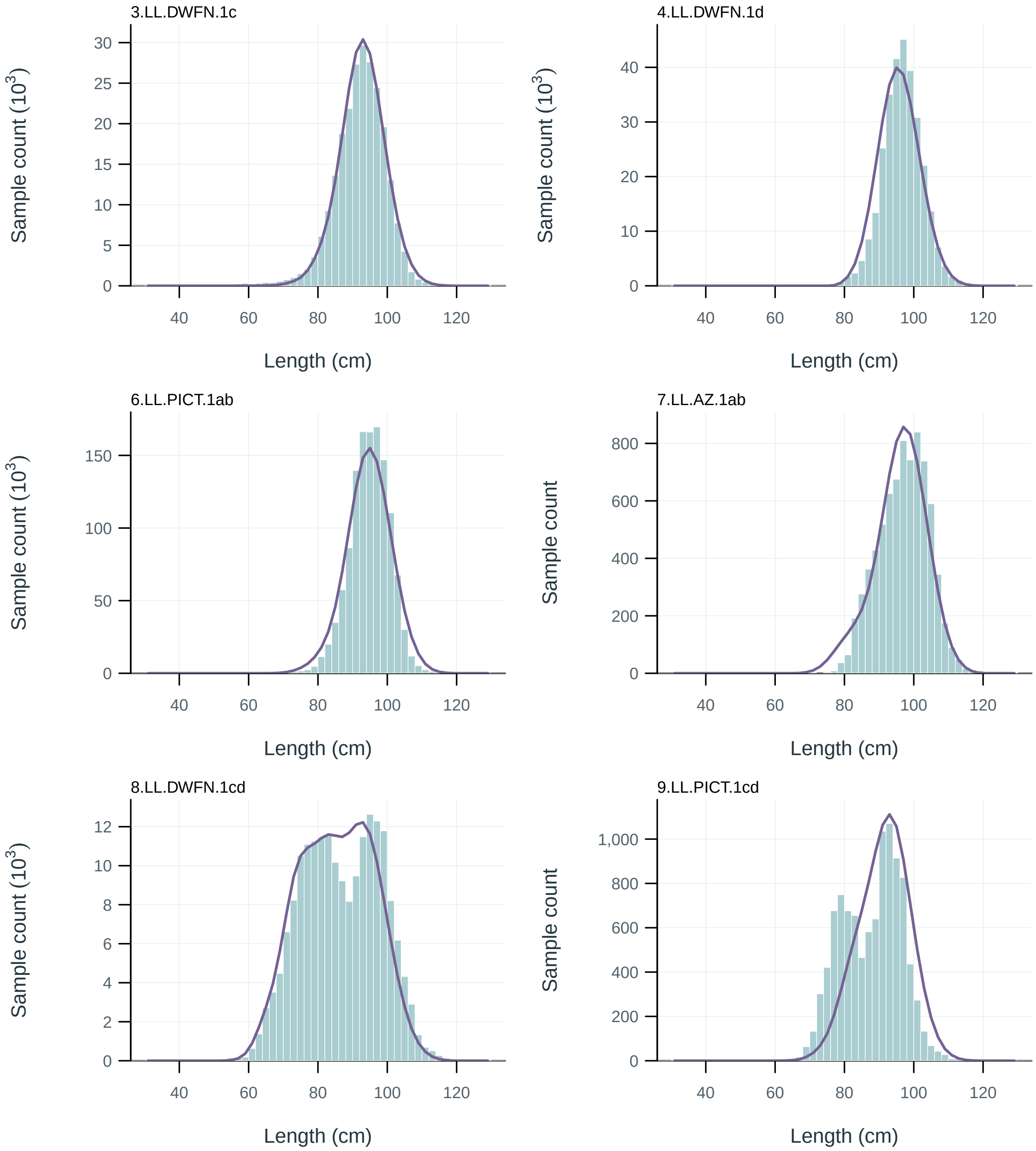

Figure 12: Length-composition fits by fishery. Observed sample counts (bars) and predictions (lines), pooled over years.

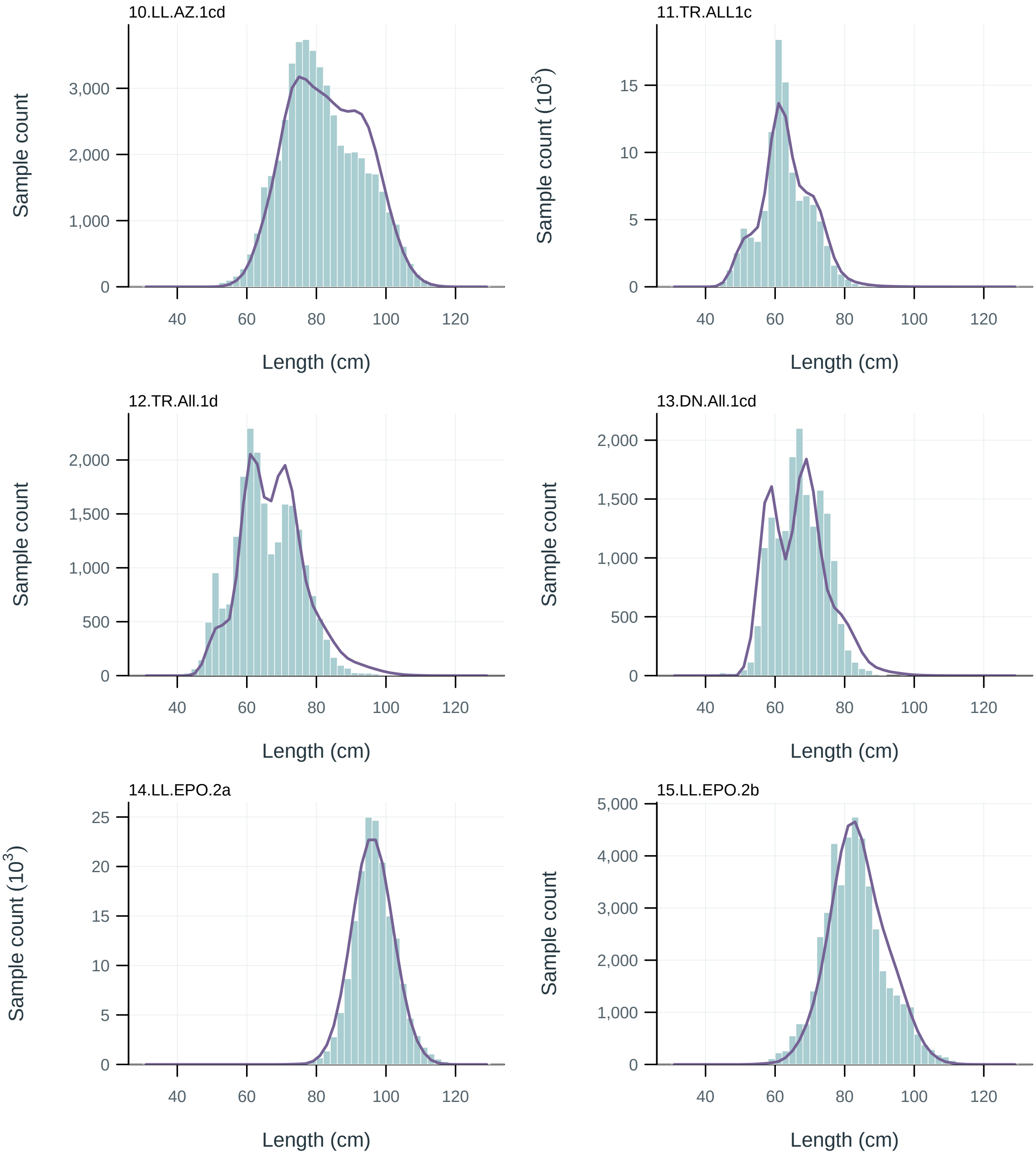

Figure 13: Length-composition fits by fishery. Observed sample counts (bars) and predictions (lines), pooled over years.

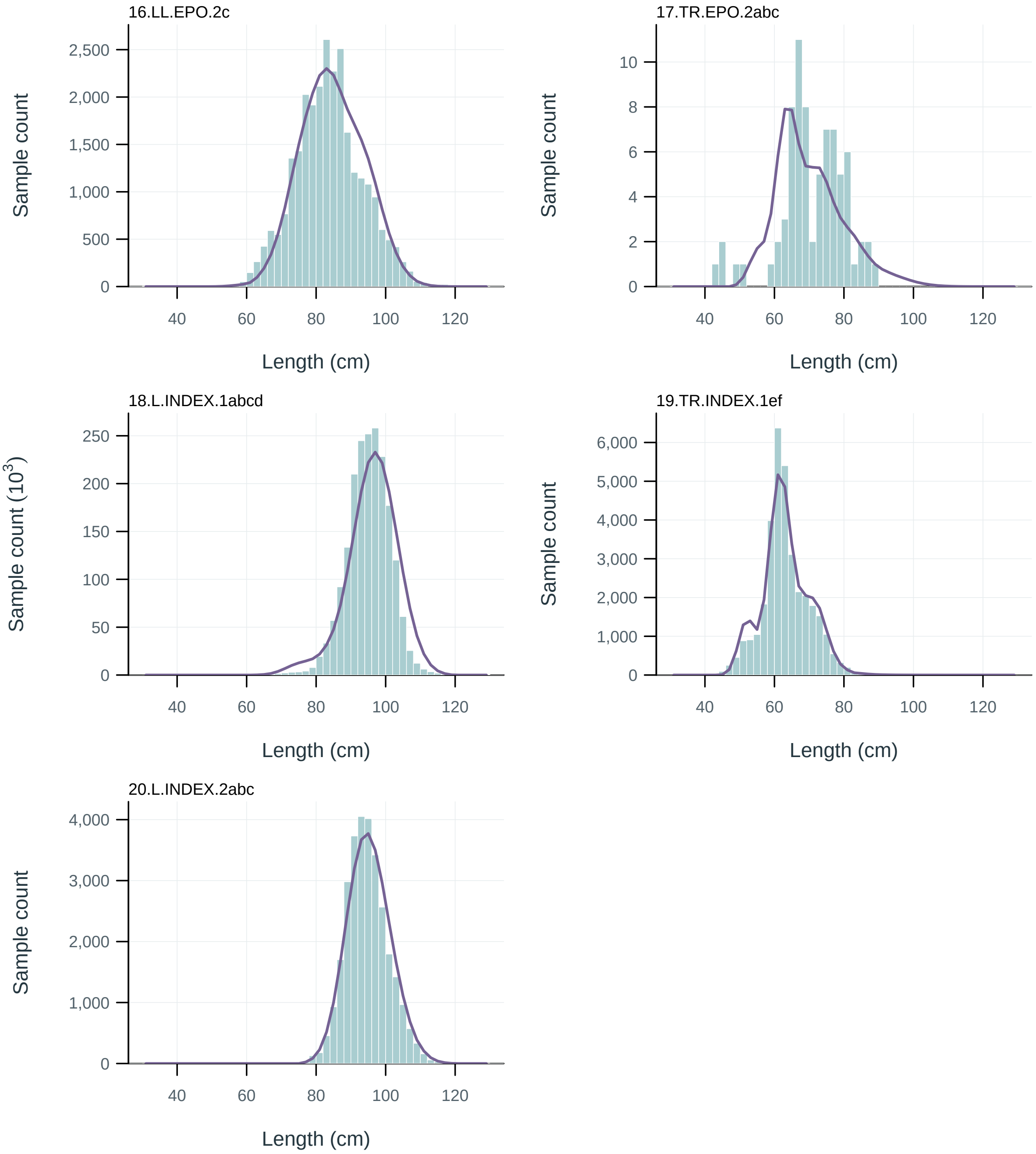

Figure 14: Length-composition fits by fishery. Observed sample counts (bars) and predictions (lines), pooled over years.

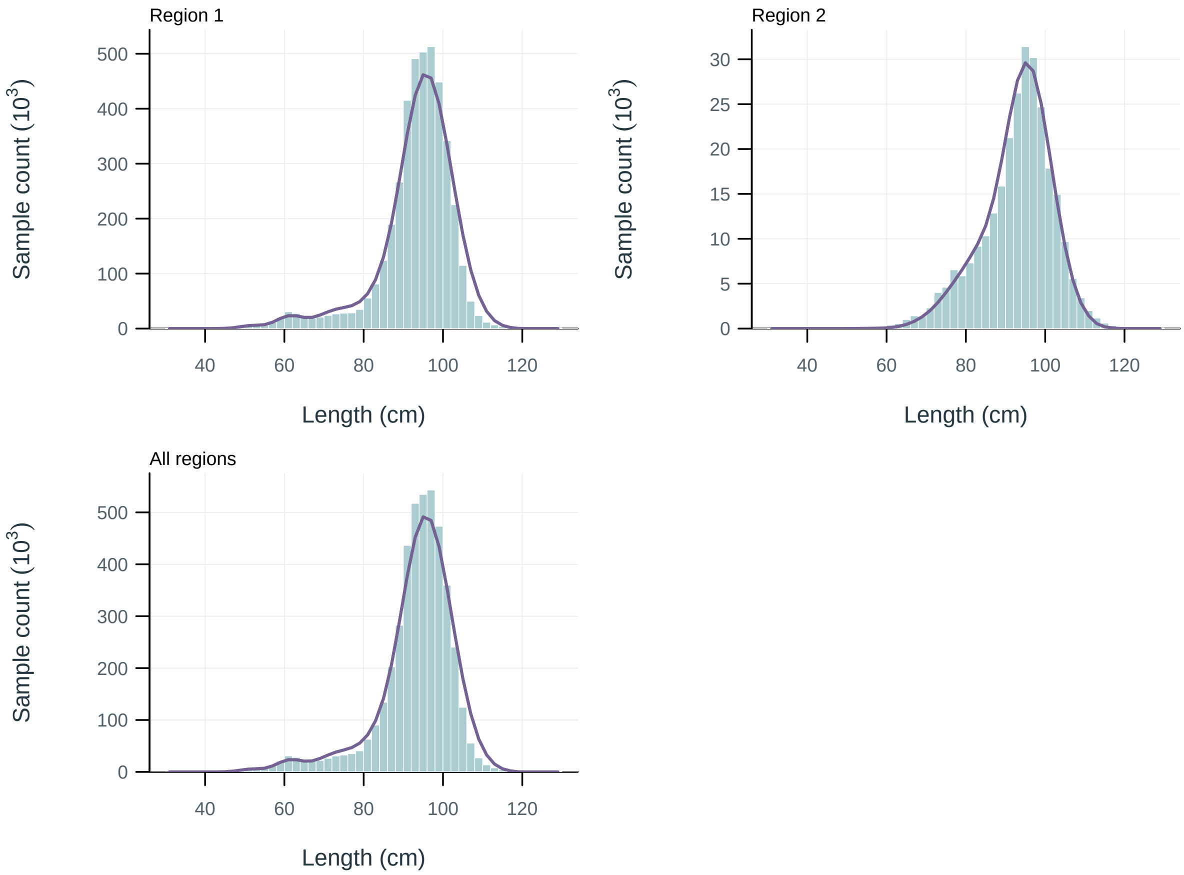

Figure 15: Length-composition fits by region. Observed sample counts (bars) and predictions (lines), pooled over years.

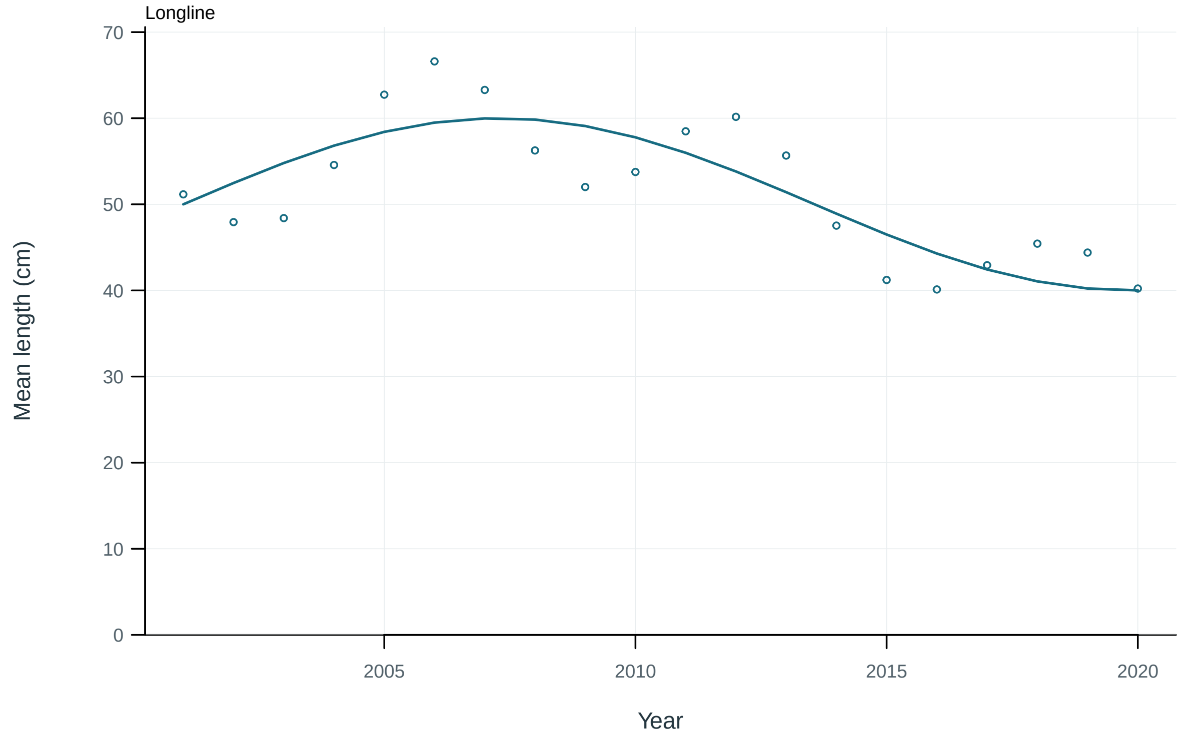

Figure 16: Illustrative output. Observed and predicted length summaries for longline fisheries.

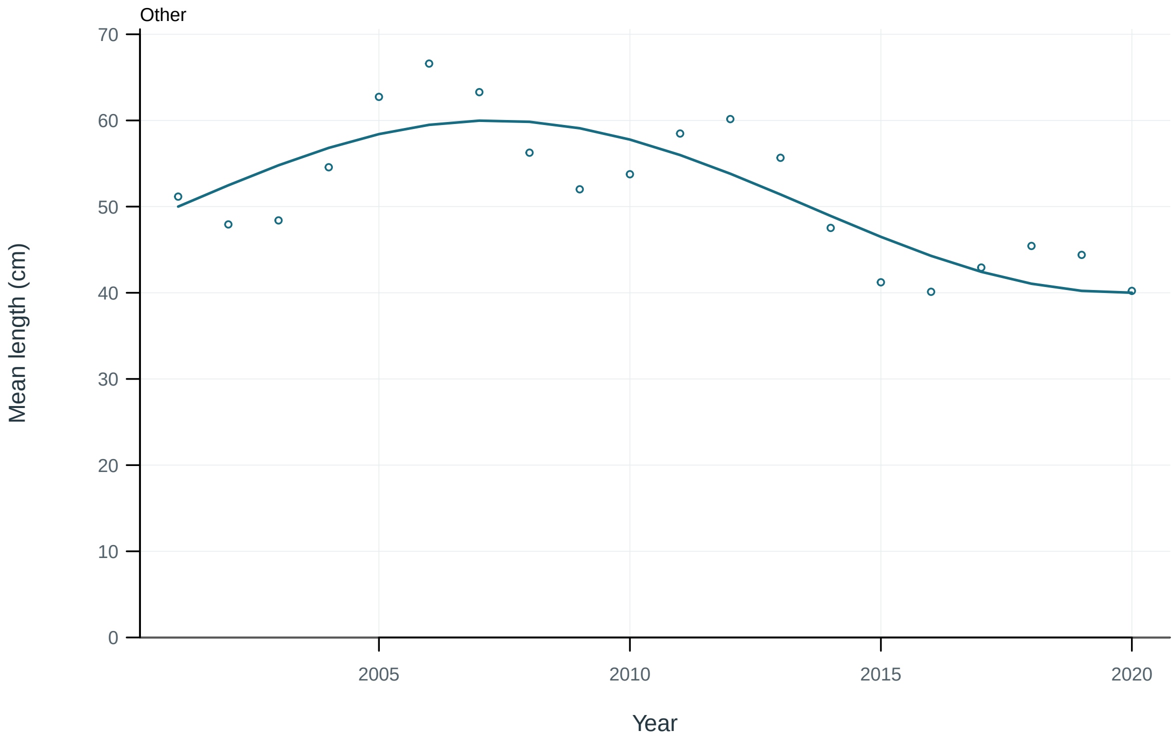

Figure 17: Illustrative output. Observed and predicted length summaries for other fisheries.

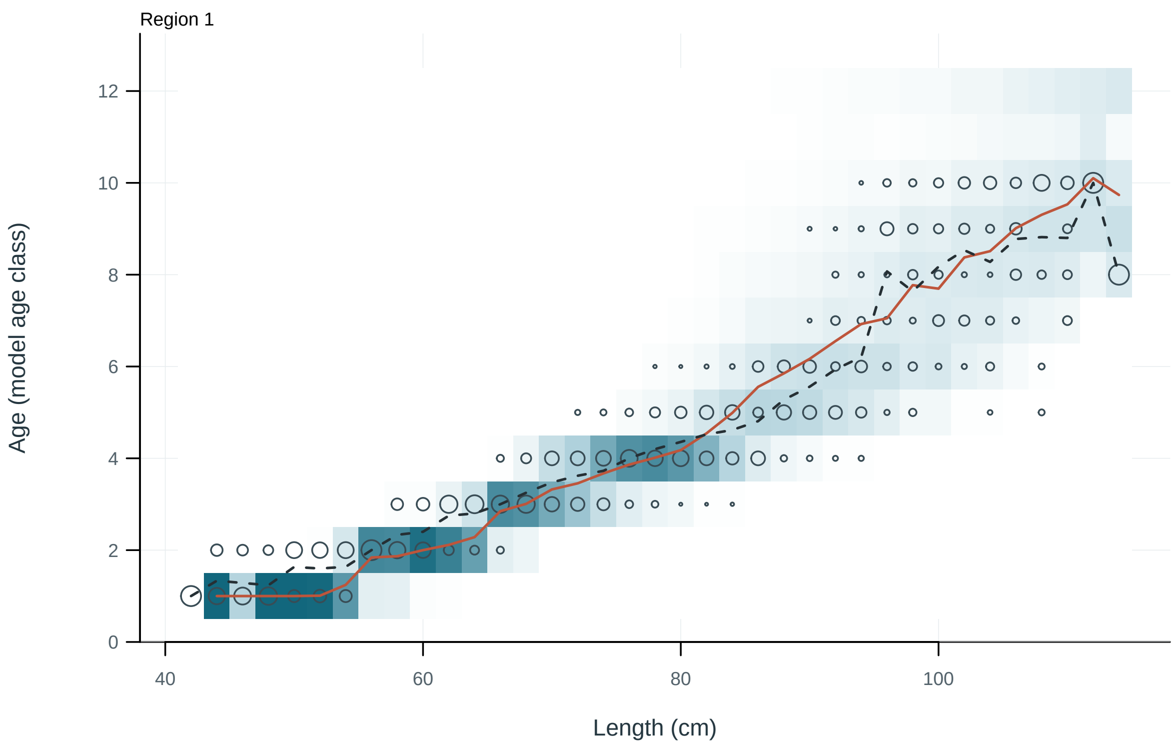

Figure 18: Observed and predicted age distributions conditional on length. Cells: predictions; circles: observations; solid and dashed lines: predicted and observed mean age.

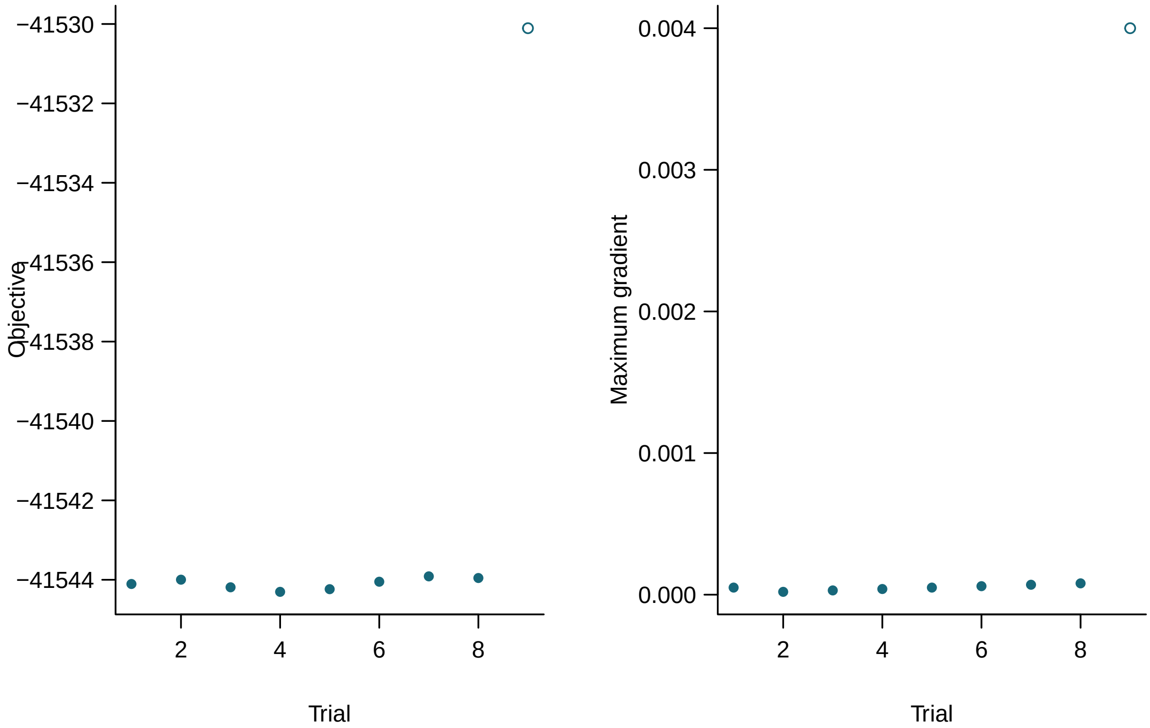

Figure 19: Final objectives and maximum gradients for independent starting values. Filled points meet all recorded numerical checks.

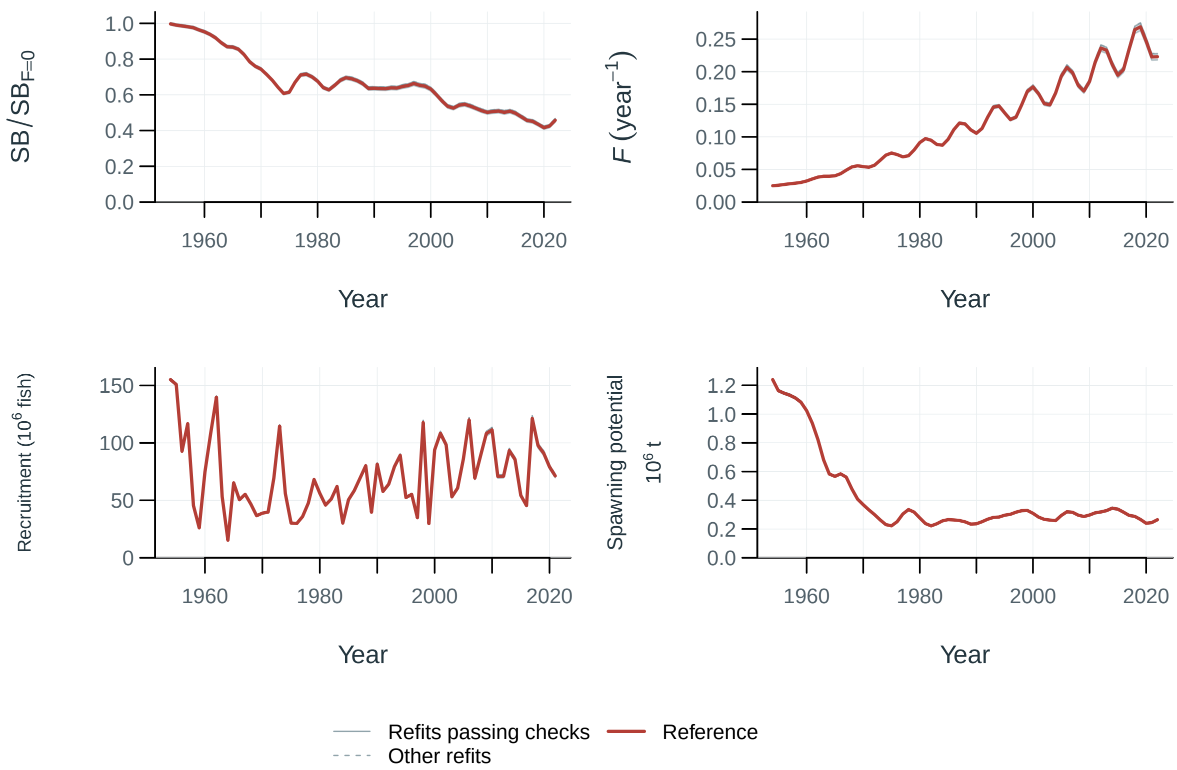

Figure 20: Jitter model trajectories.

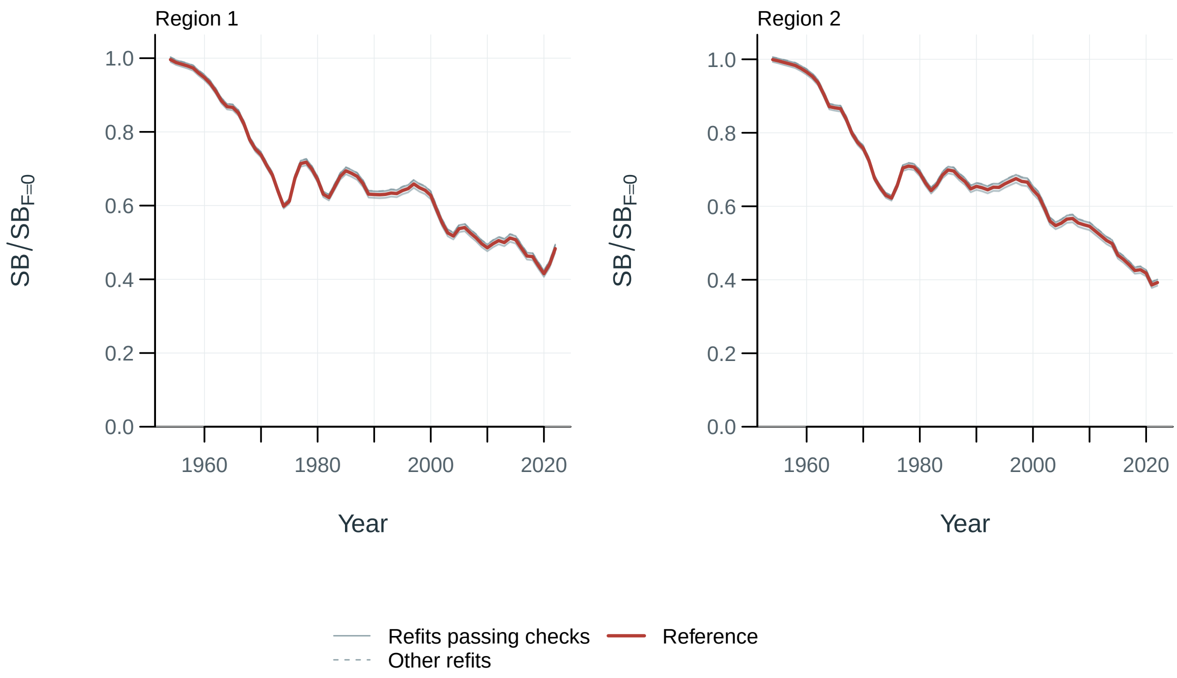

Figure 21: Spawning depletion by model and region, relative to each model's dynamic unfished reference.

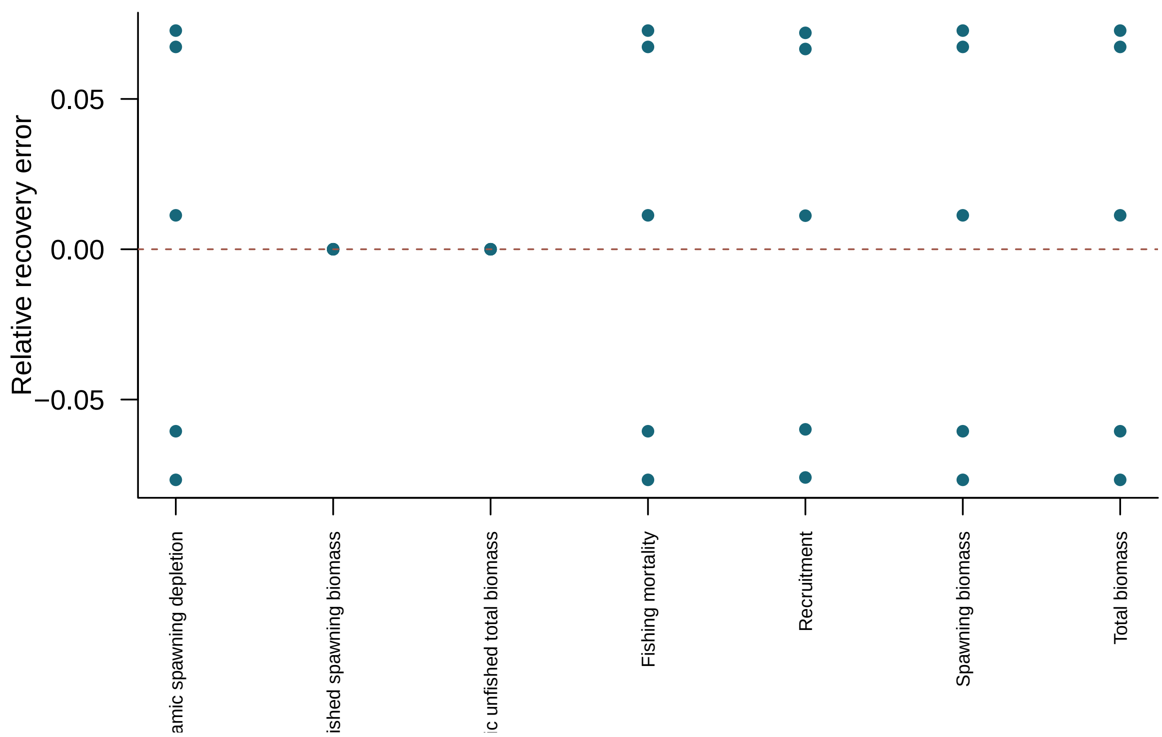

Figure 22: Relative recovery error in terminal quantities for converged simulated-data refits.

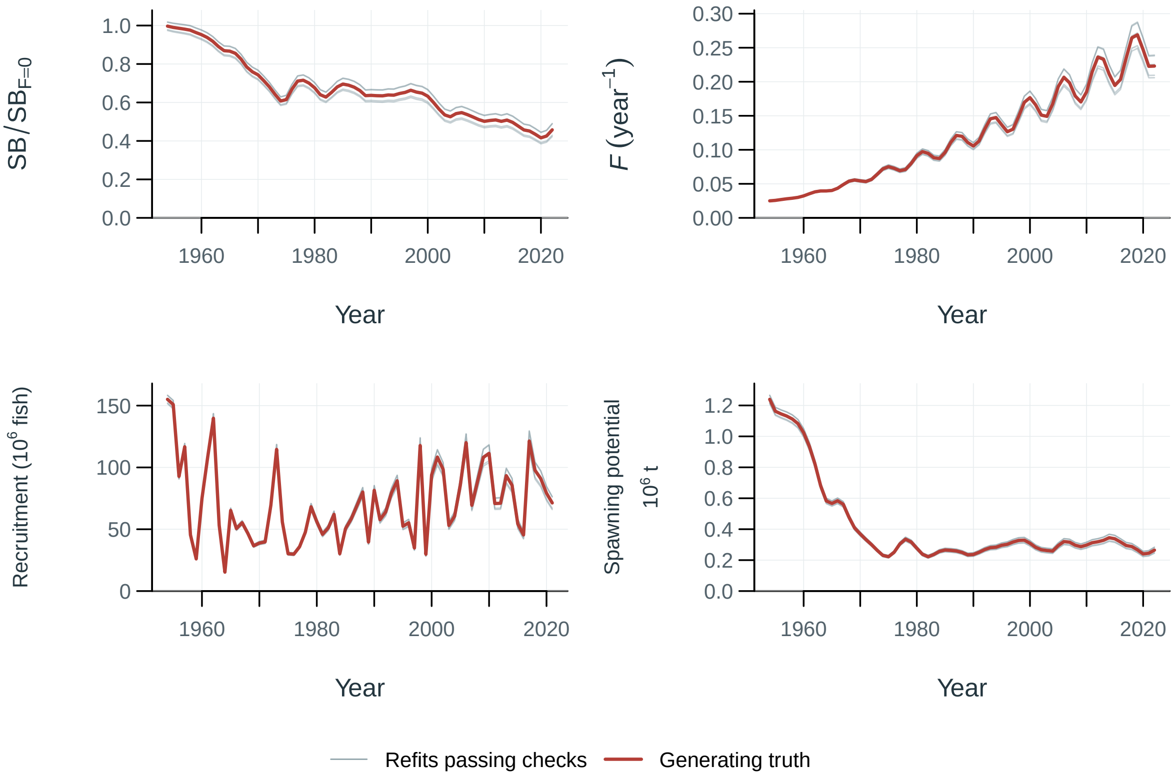

Figure 23: Selftest model trajectories.

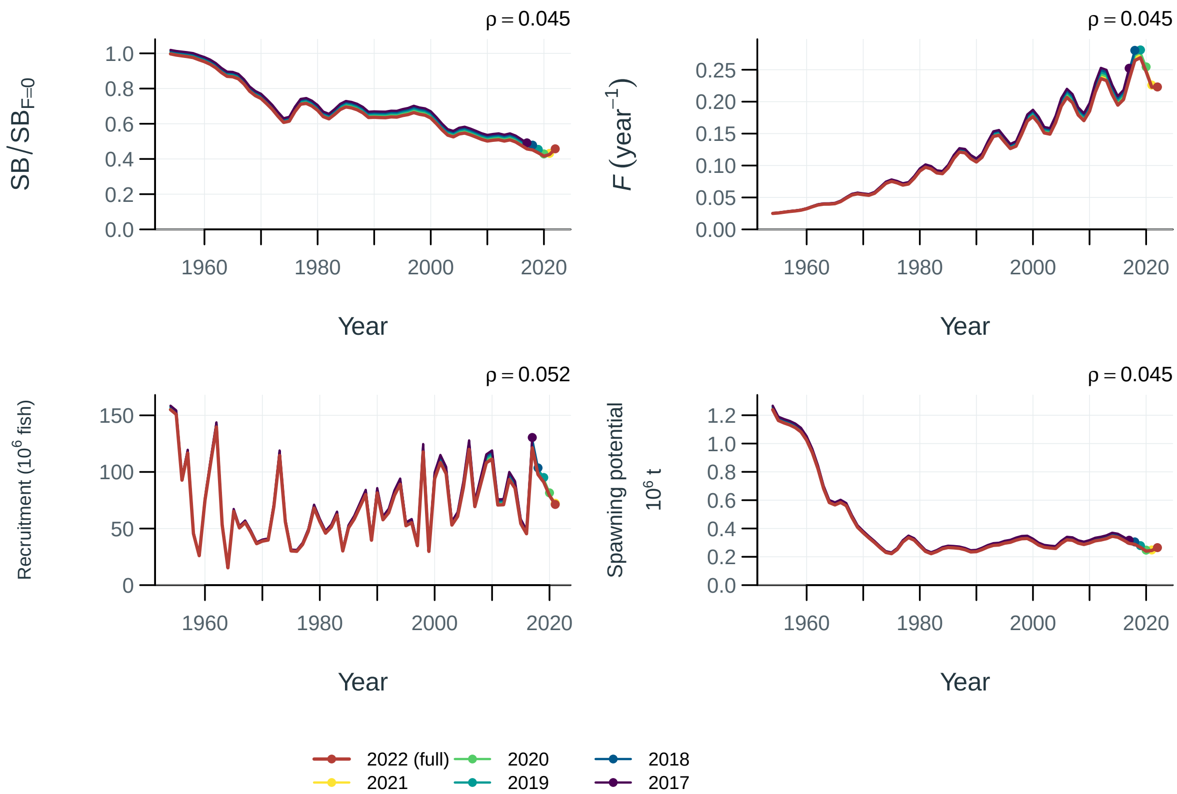

Figure 24: Population trajectories for the full-data fit and retrospective peels. Points mark terminal years; rho is Mohn's mean terminal relative difference for peels passing the recorded numerical checks.

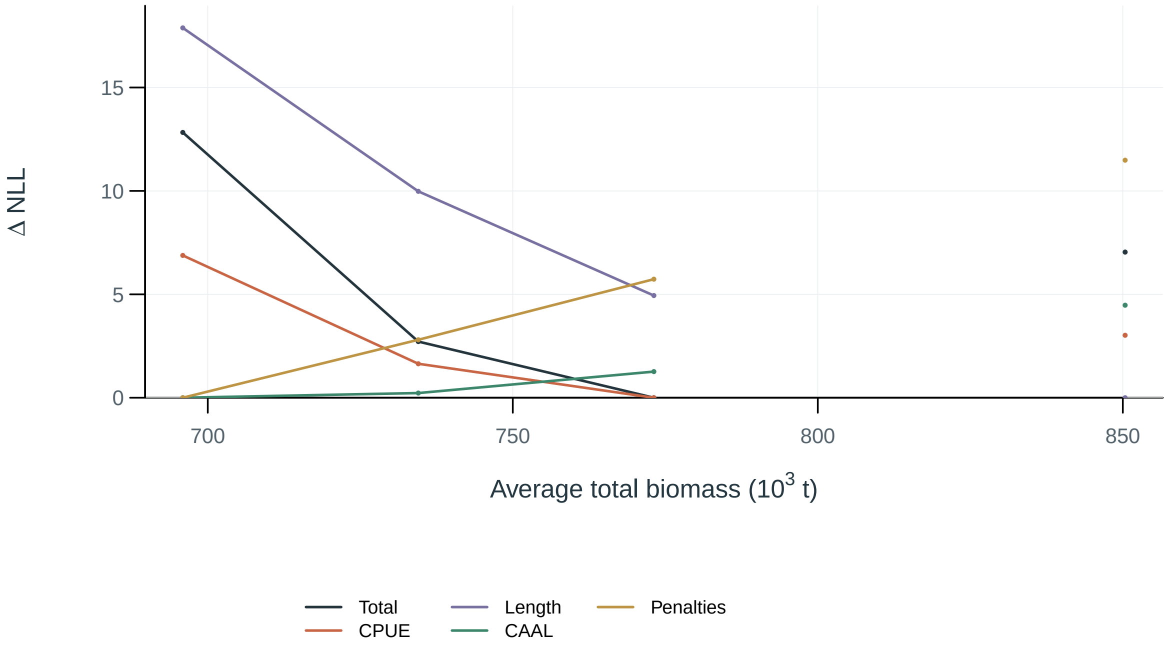

Figure 25: Total negative log-likelihood and component profiles. Each curve is centered on its own minimum. Total biomass summed over regions and averaged across model periods in 1954-2022.

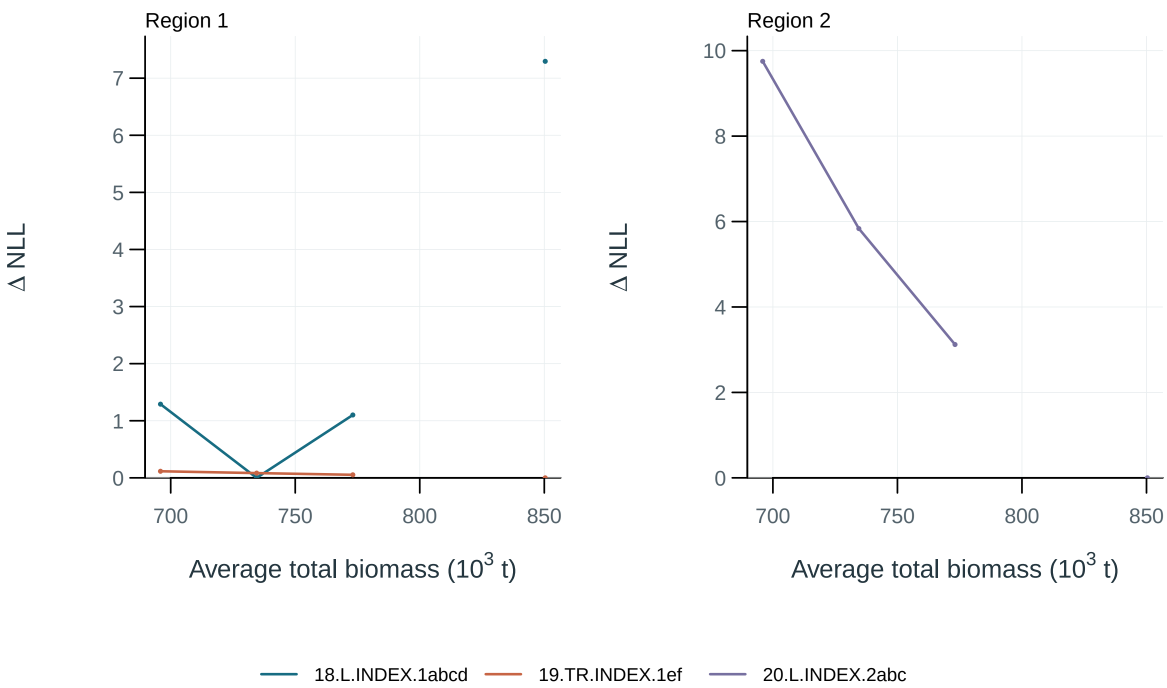

Figure 26: Negative log-likelihood profiles for abundance-index components. Each curve is centered on its own minimum. Total biomass summed over regions and averaged across model periods in 1954-2022.

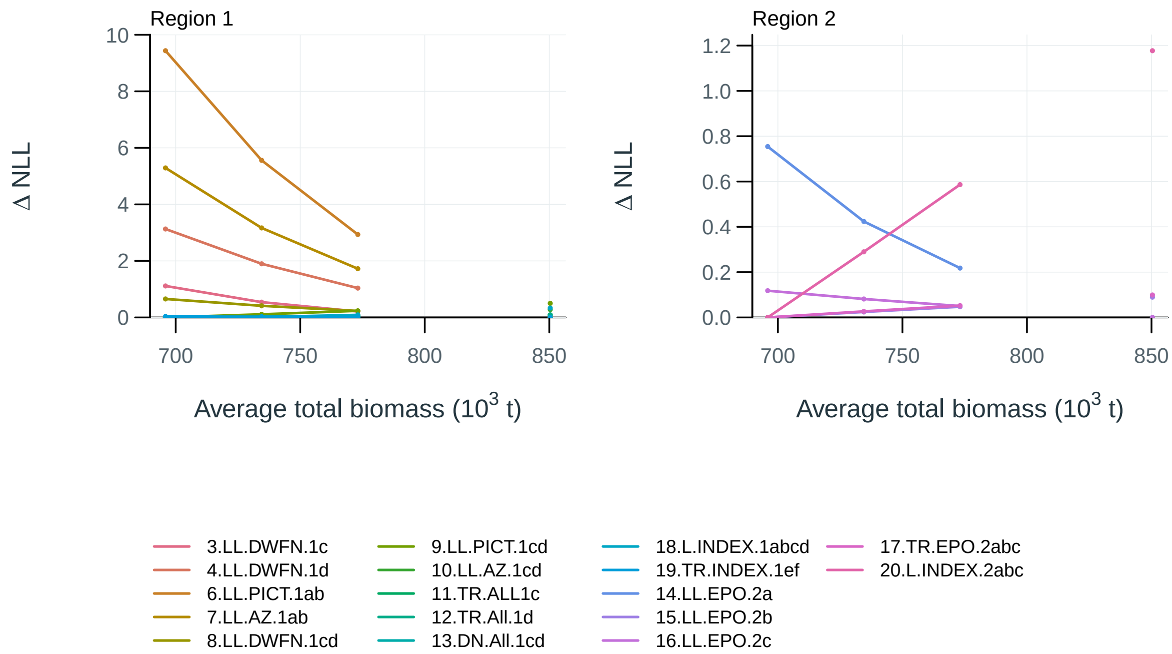

Figure 27: Negative log-likelihood profiles for length-composition components. Each curve is centered on its own minimum. Total biomass summed over regions and averaged across model periods in 1954-2022.

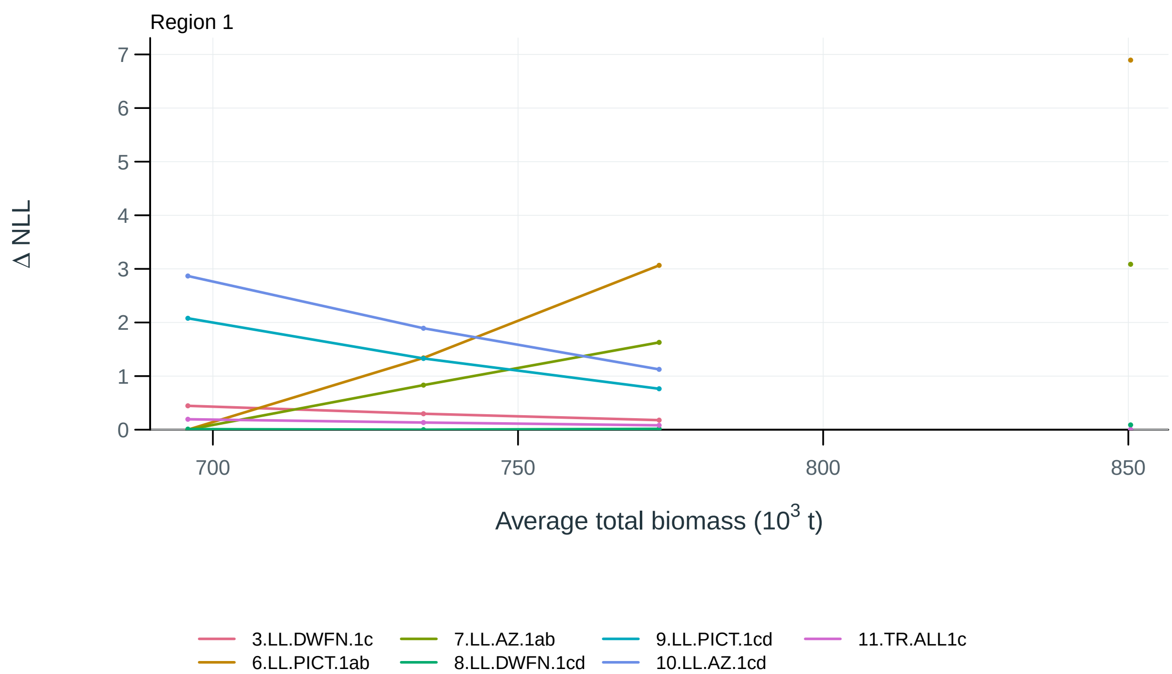

Figure 28: Negative log-likelihood profiles for conditional age-at-length components. Each curve is centered on its own minimum. Total biomass summed over regions and averaged across model periods in 1954-2022.

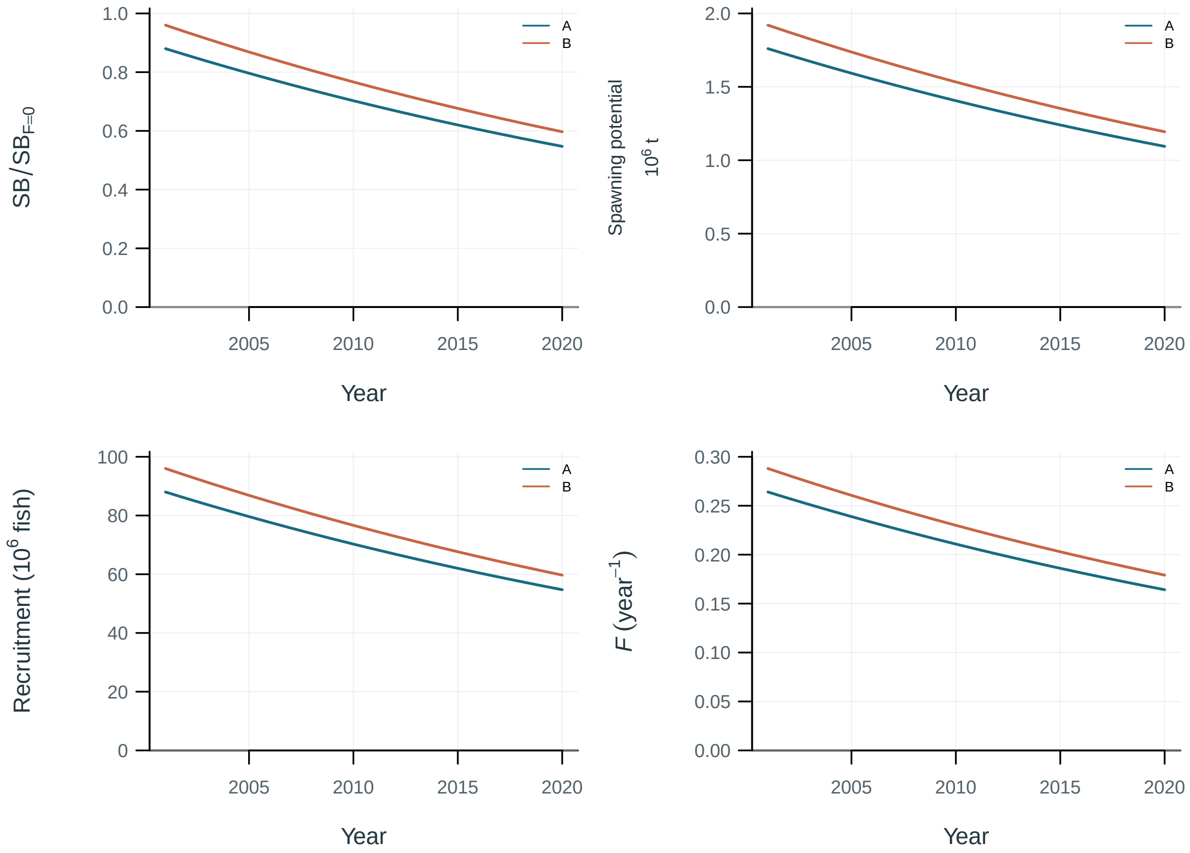

Figure 29: Illustrative output. Diagnostic and production-model trajectories.

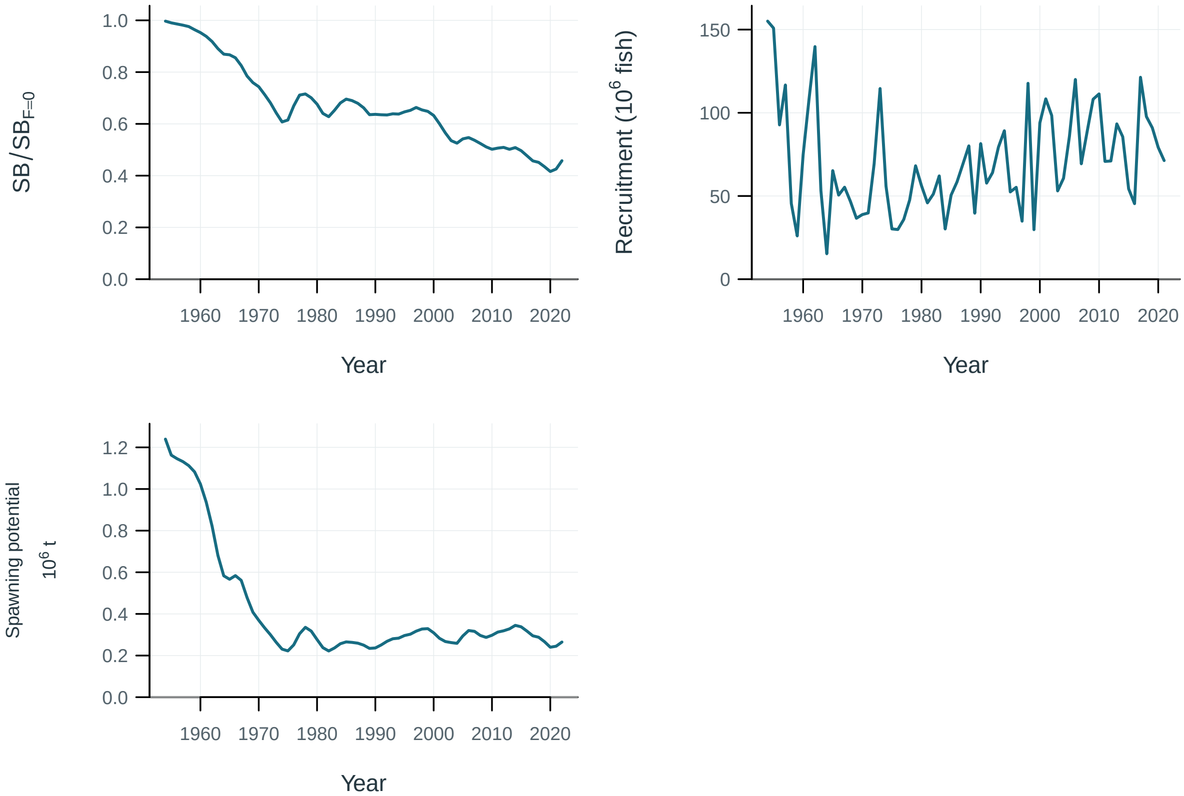

Figure 30: Estimated population trajectories.

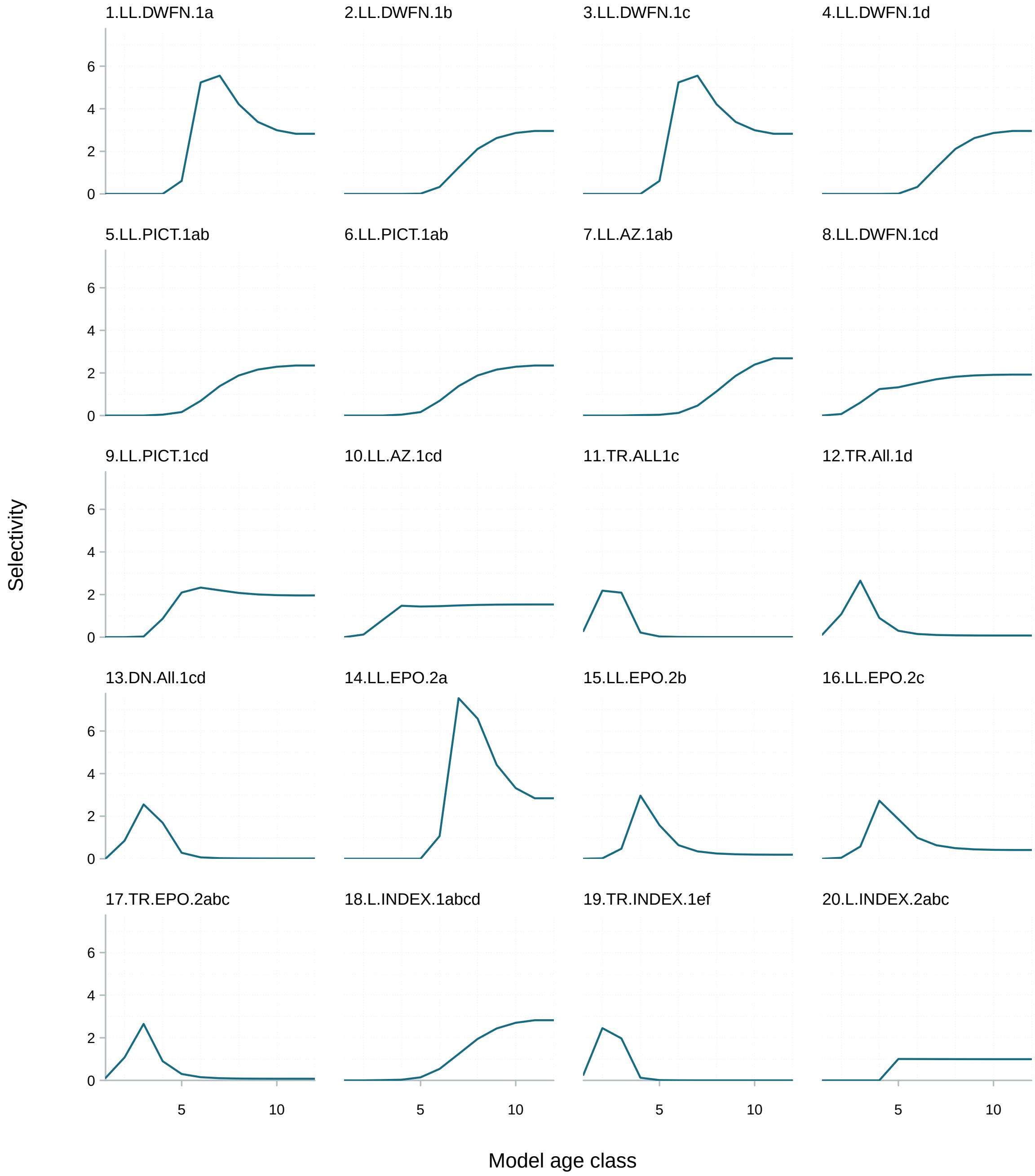

Figure 31: Fishery selectivity at age.

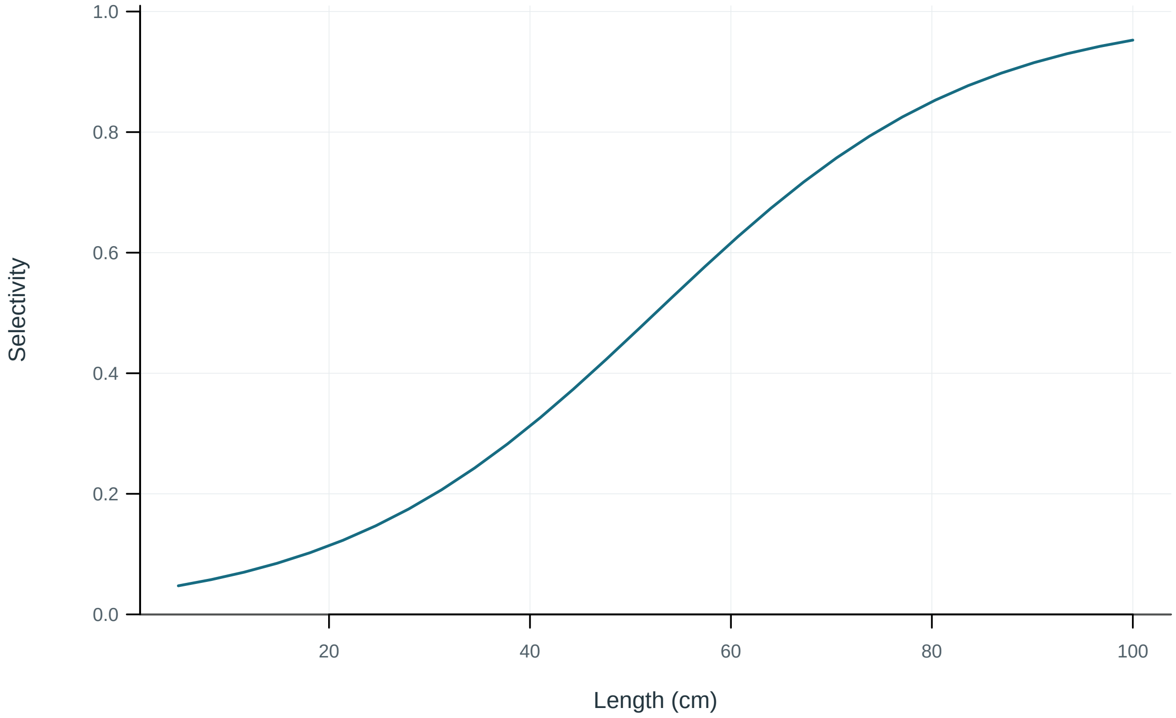

Figure 32: Illustrative output. Selectivity at length by fishery.

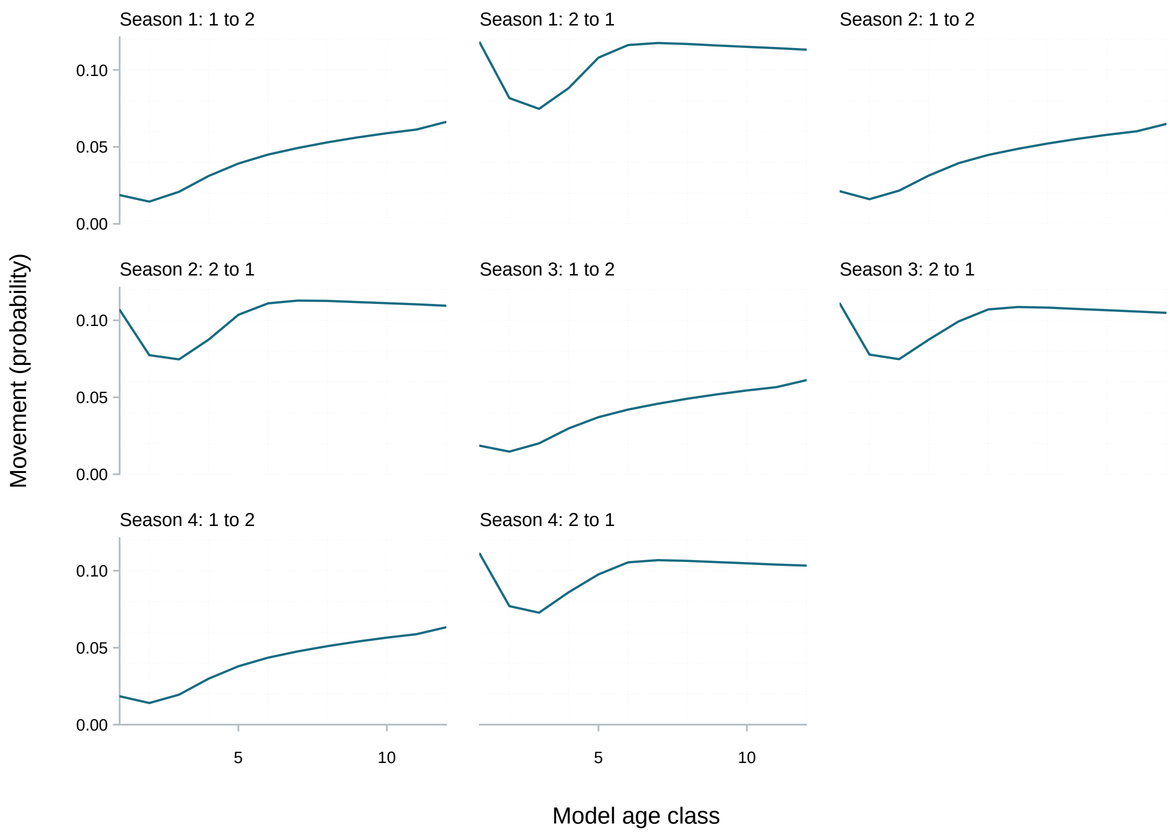

Figure 33: Probability of movement from source to destination region per movement event.

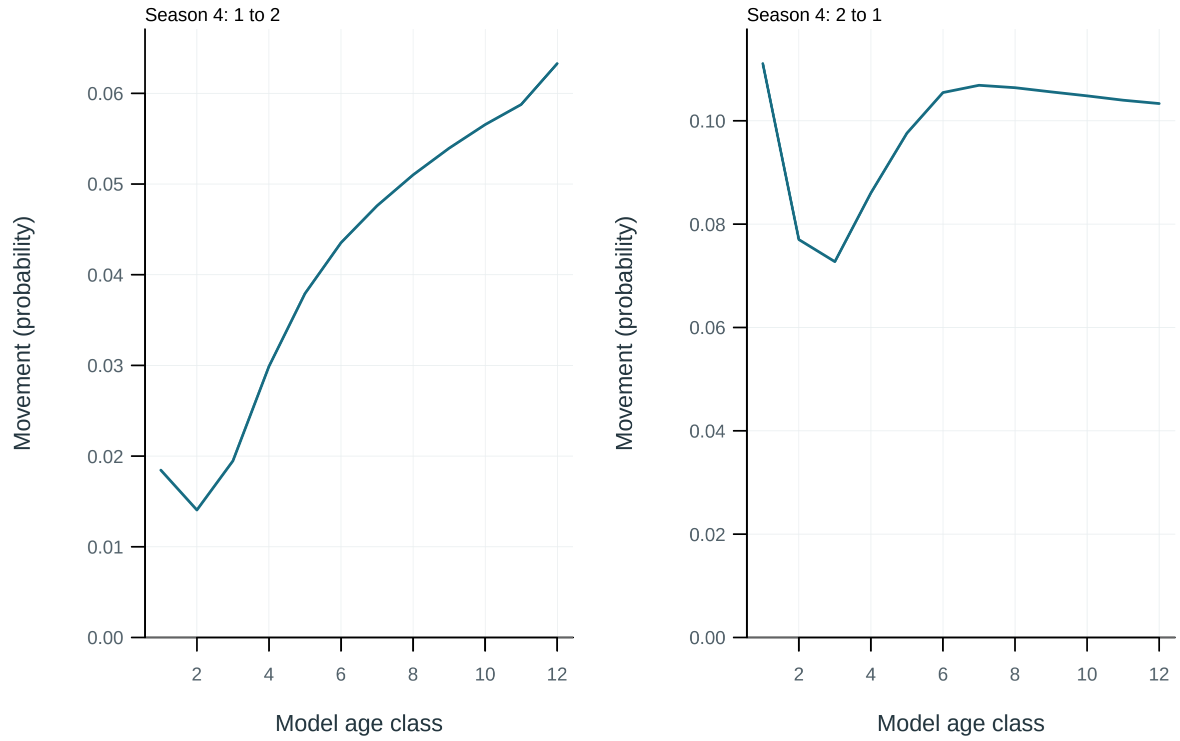

Figure 34: Probability of movement from source to destination region per movement event.

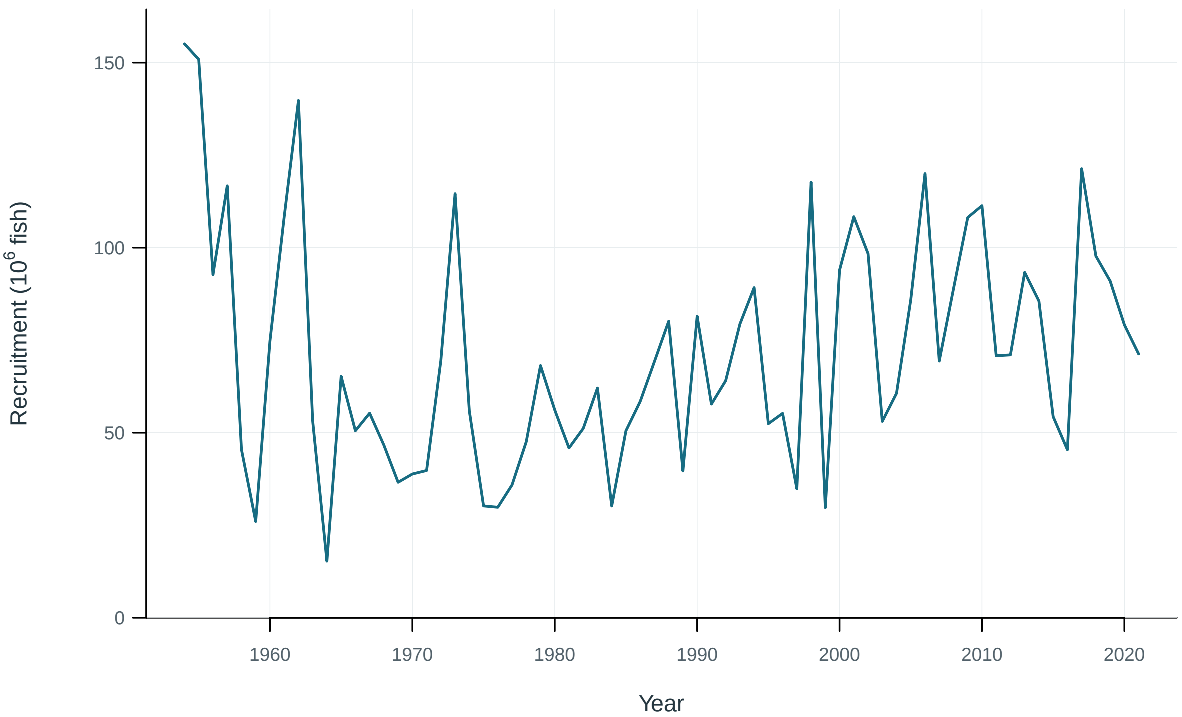

Figure 35: Annual recruitment.

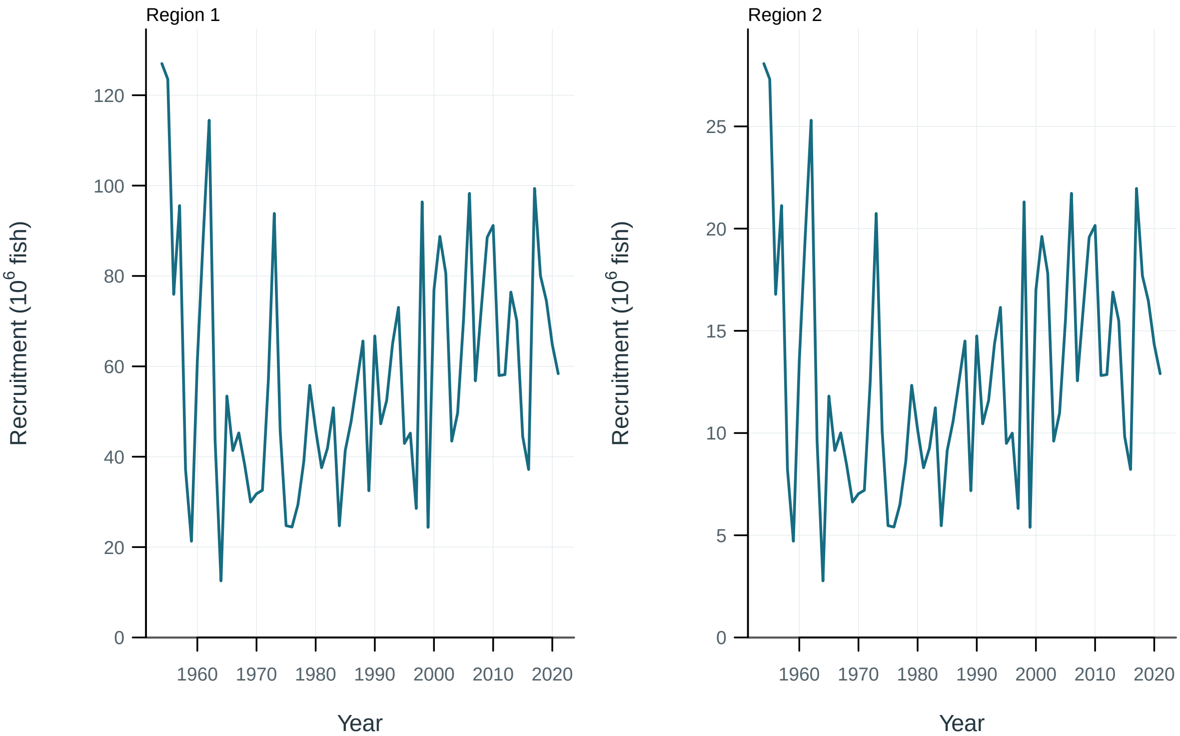

Figure 36: Recruitment by model region.

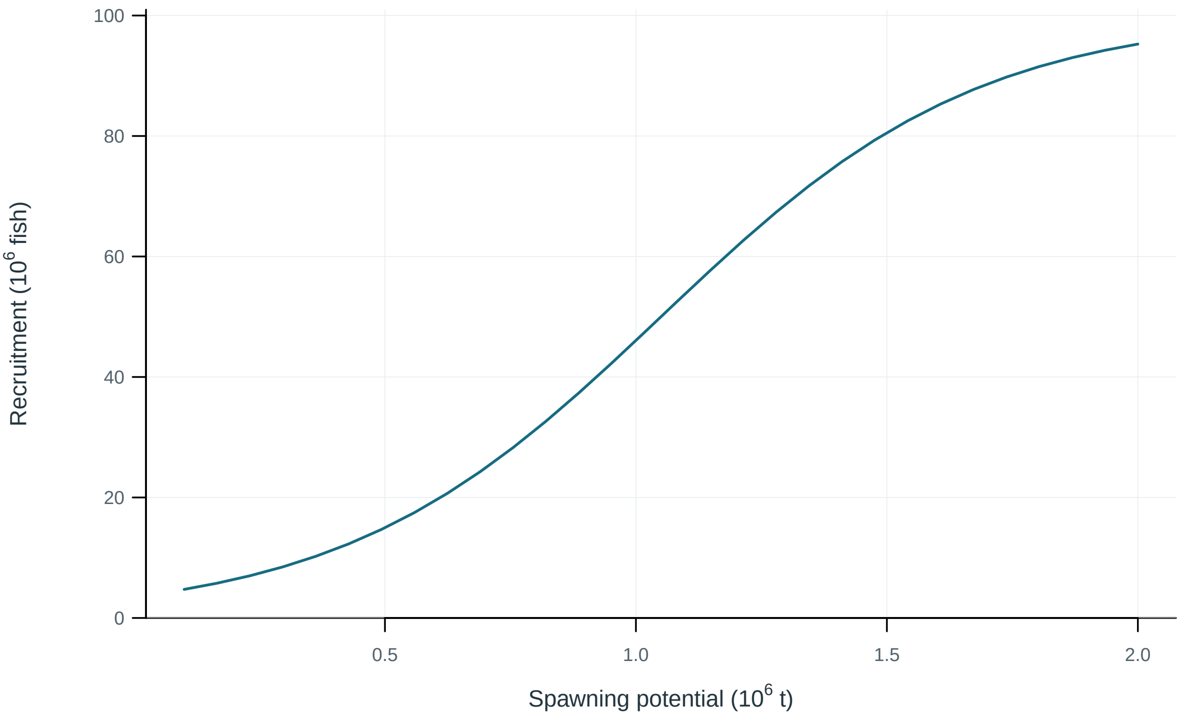

Figure 37: Illustrative output. Stock–recruitment observations and fitted relationship.

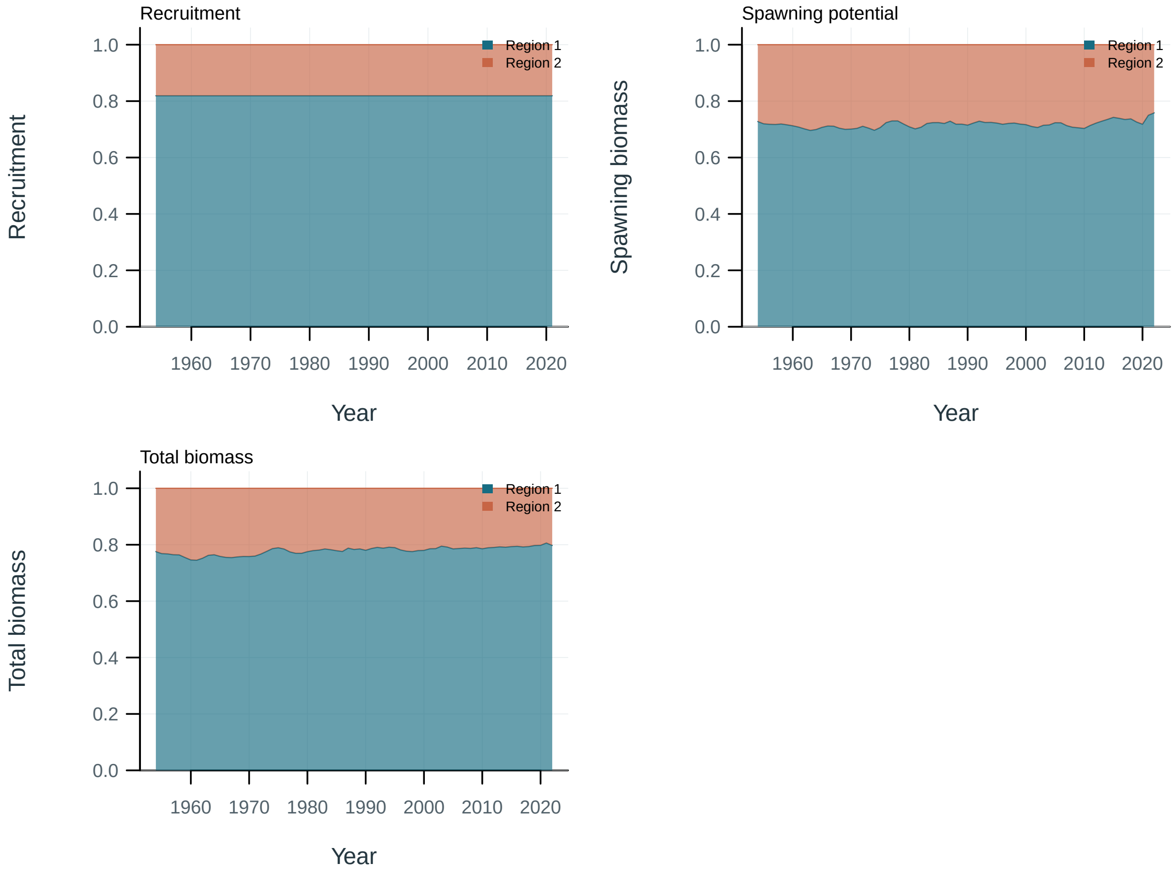

Figure 38: Regional shares of population quantities.

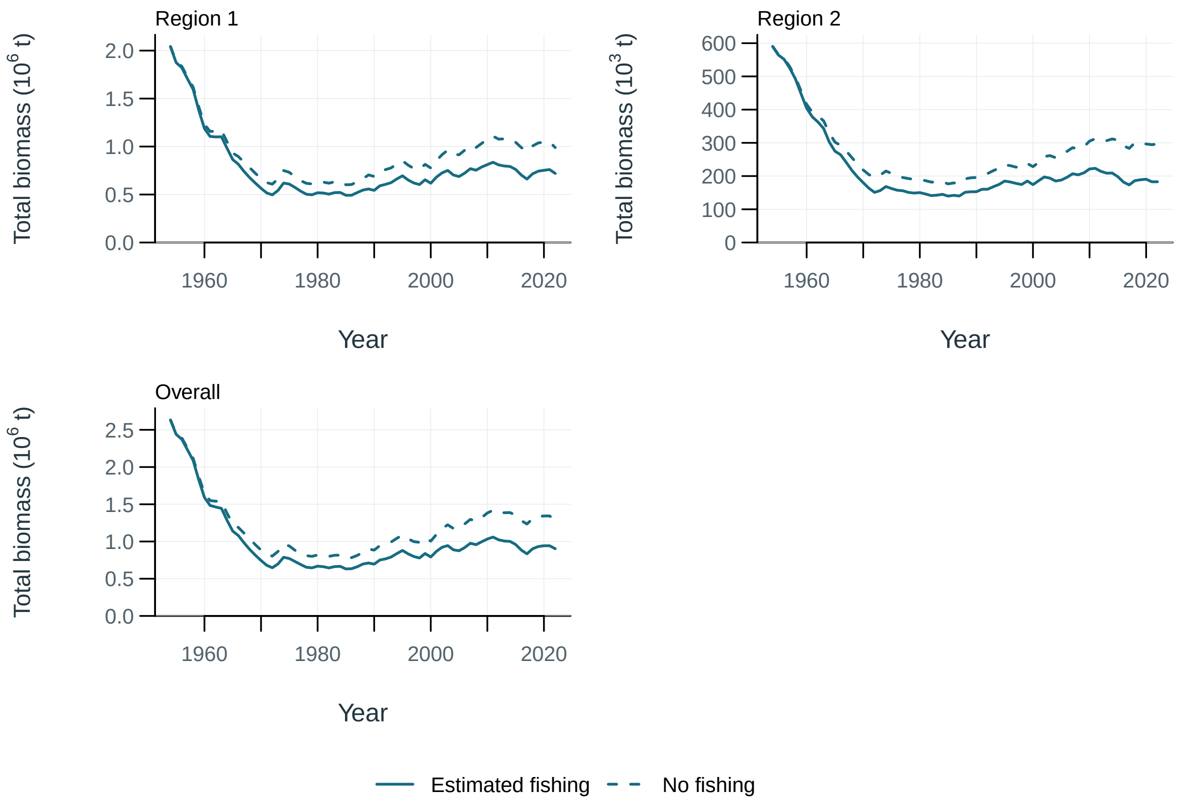

Figure 39: Total biomass under estimated fishing (solid) and the dynamic no-fishing counterfactual (dashed), by region and overall.

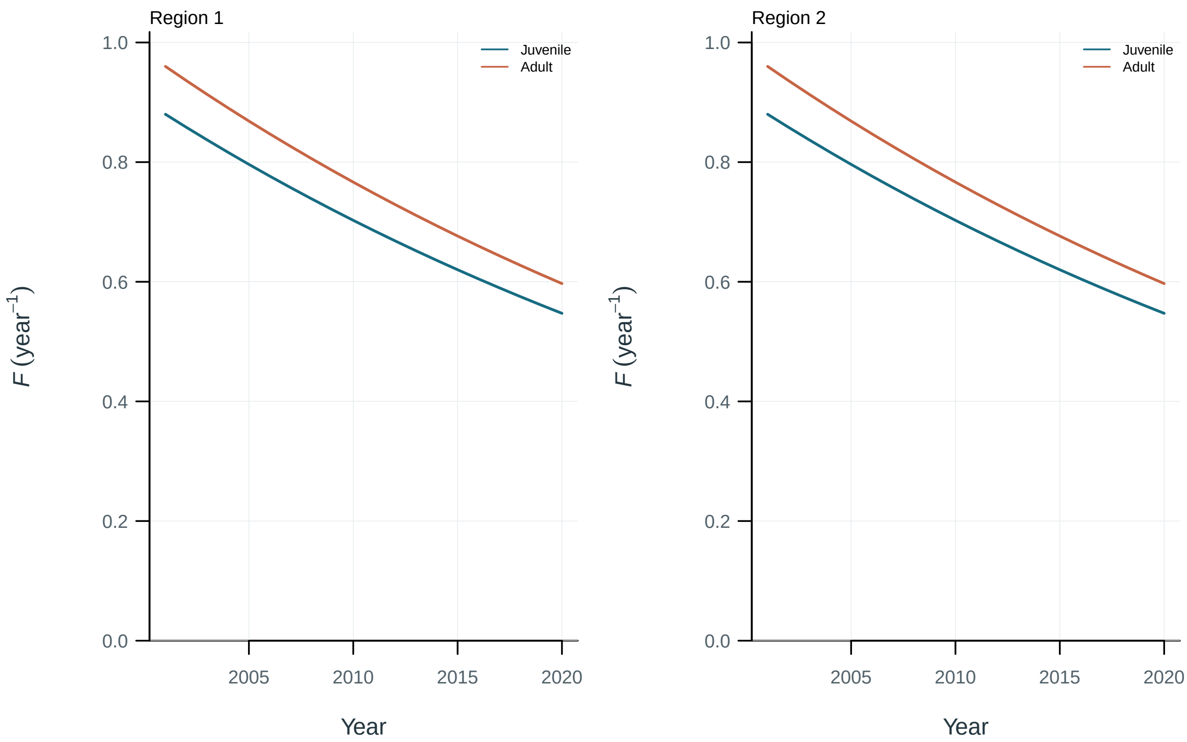

Figure 40: Illustrative output. Fishing mortality by life stage and region.

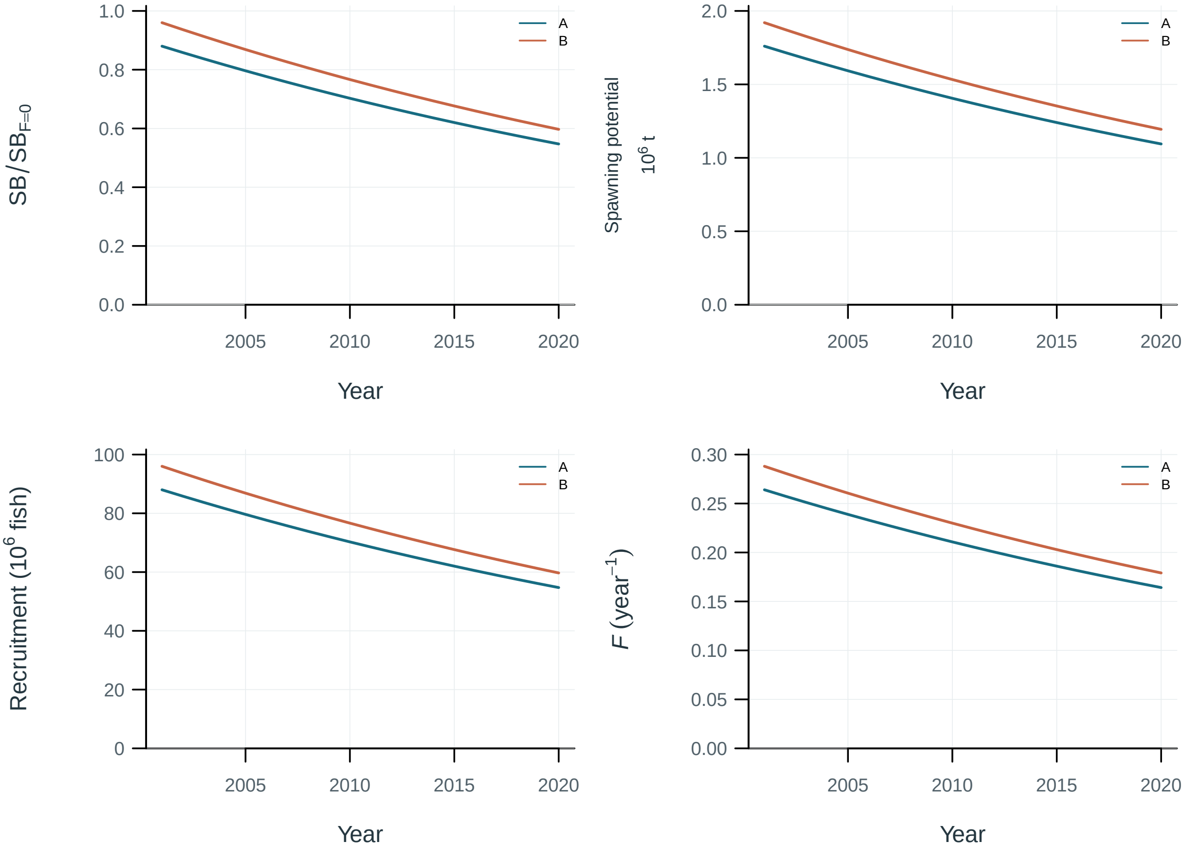

Figure 41: Sensitivity of population trajectories to steepness.

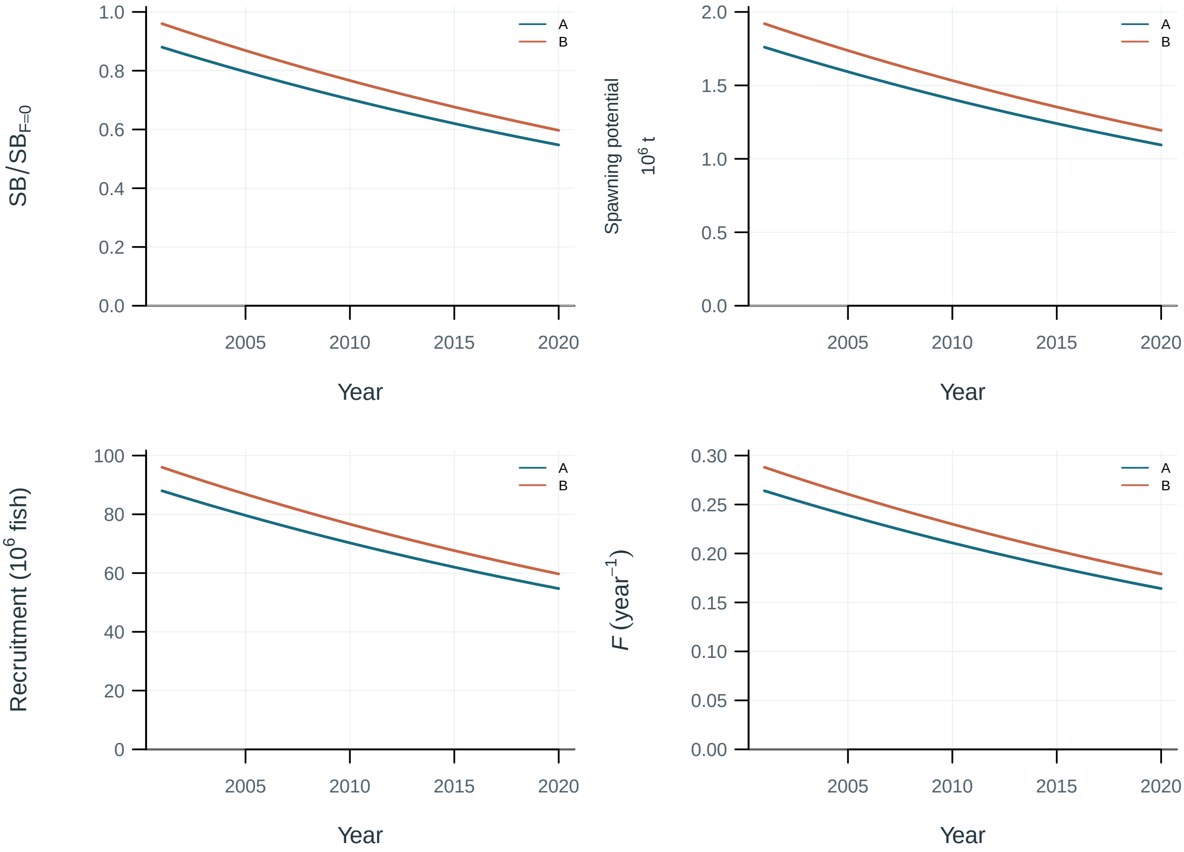

Figure 42: Sensitivity of population trajectories to natural mortality.

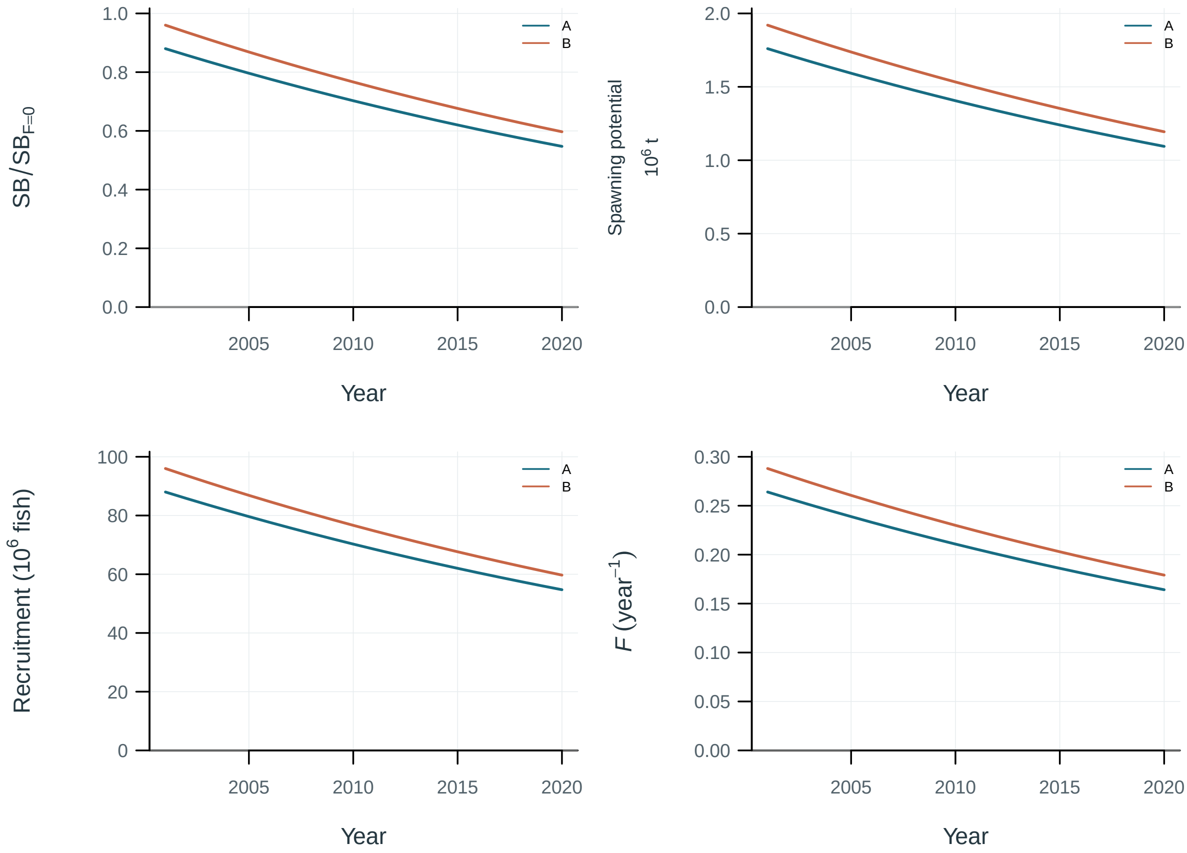

Figure 43: Sensitivity of population trajectories to regional scaling.

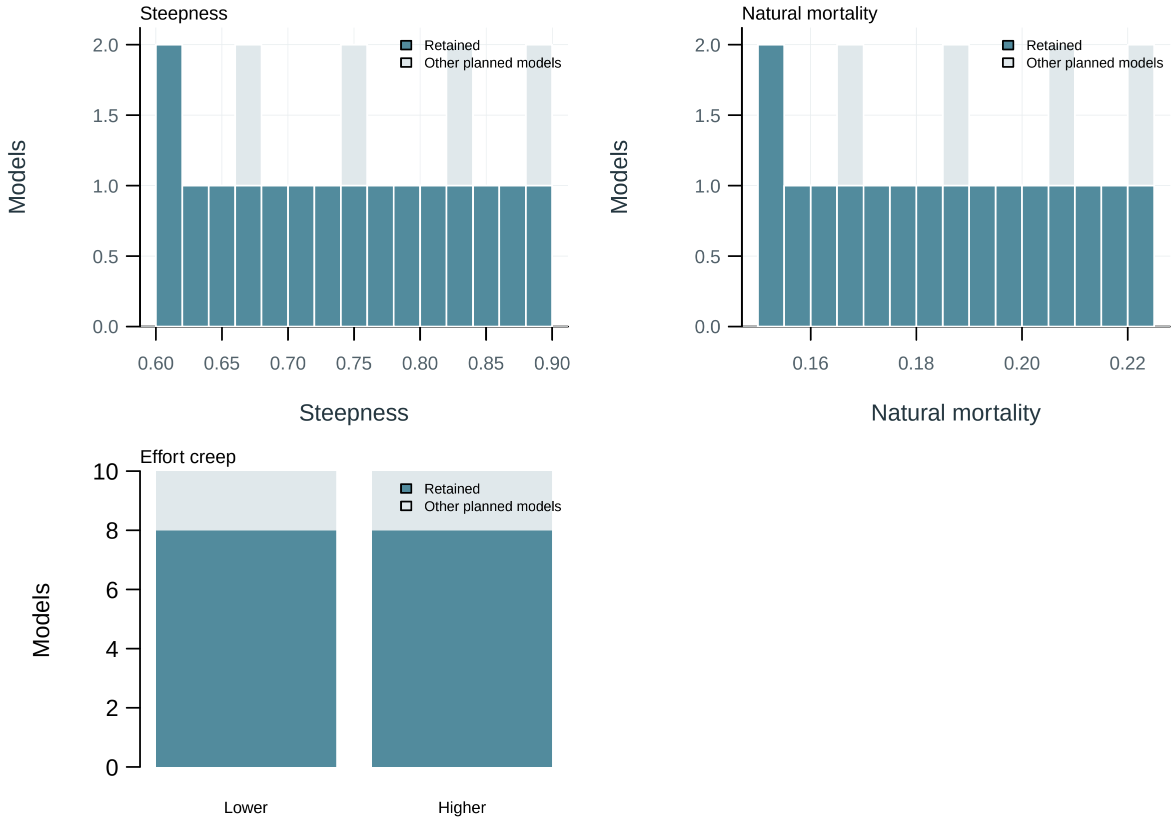

Figure 44: Planned and retained model inputs.

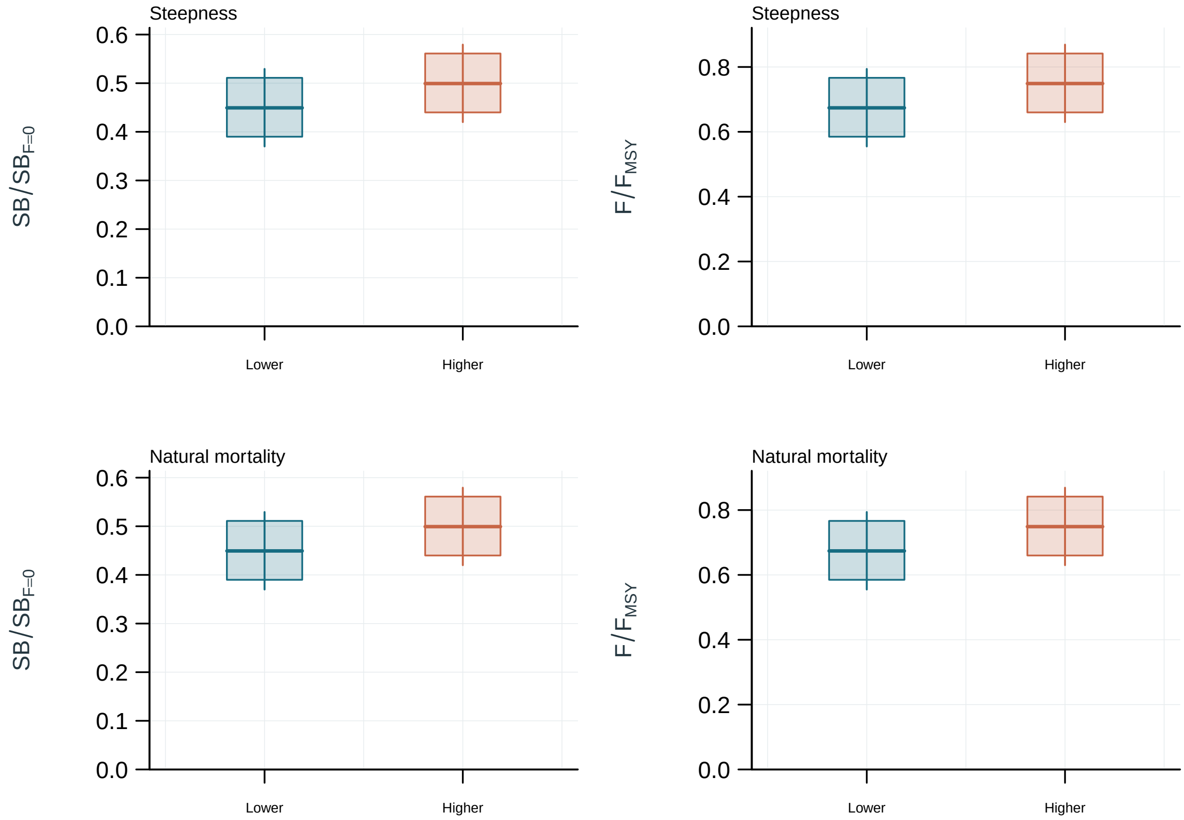

Figure 45: Weighted distributions of management quantities by design setting. Boxes: interquartile range; line: median; whiskers: 2.5th and 97.5th weighted percentiles.

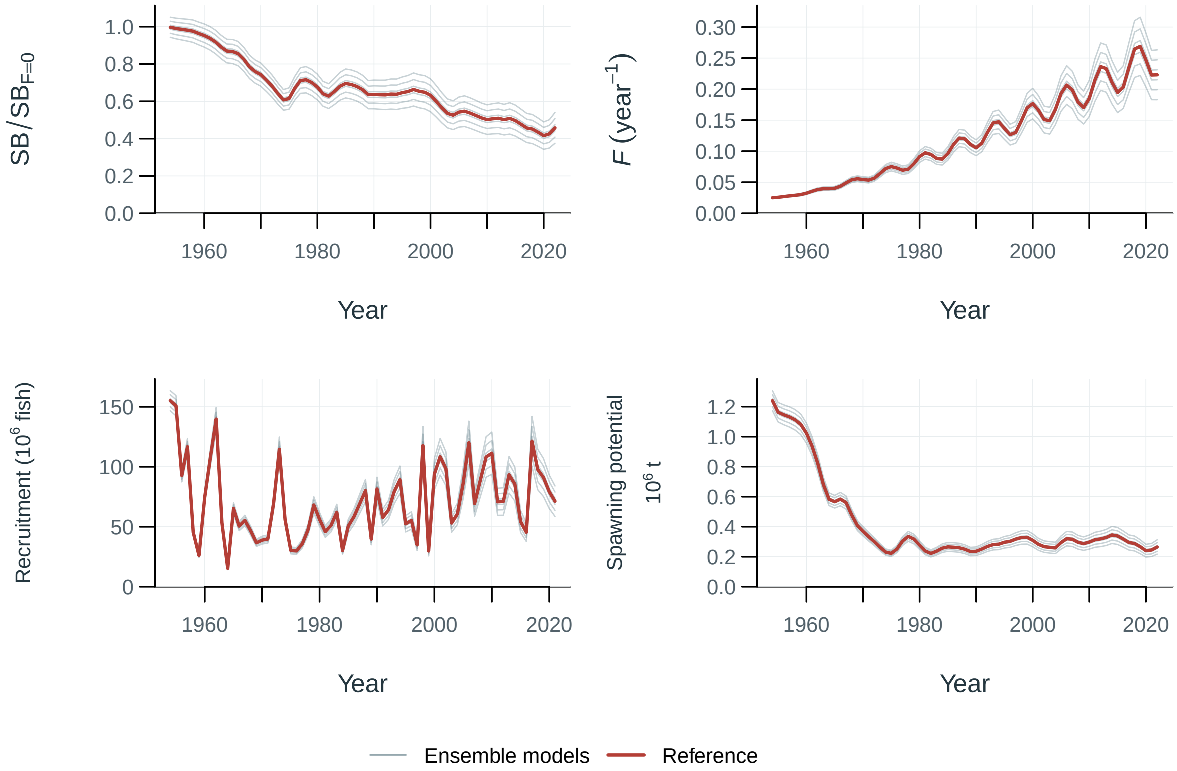

Figure 46: Population trajectories across ensemble models (thin lines) and the reference model (red).

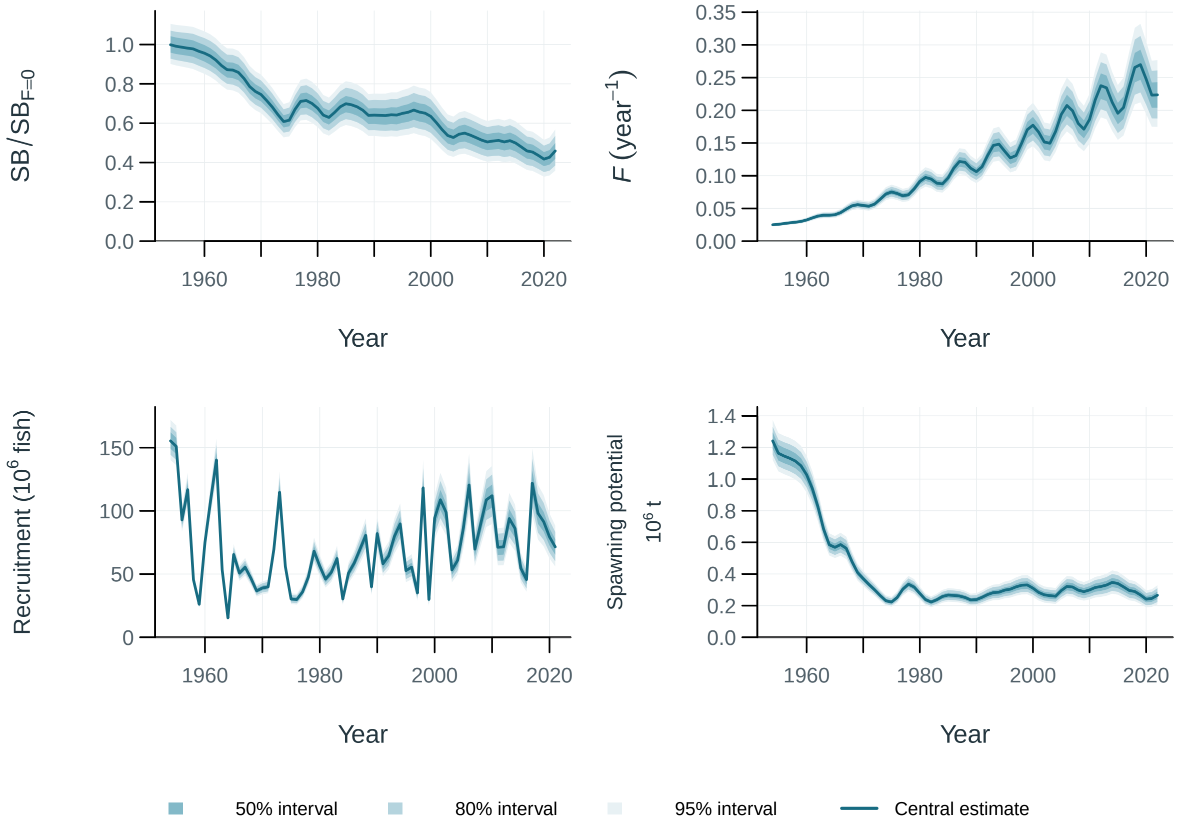

Figure 47: Population trajectories with supplied uncertainty intervals. Central 50, 80, 95% intervals (illustrative model spread and parameter perturbations).

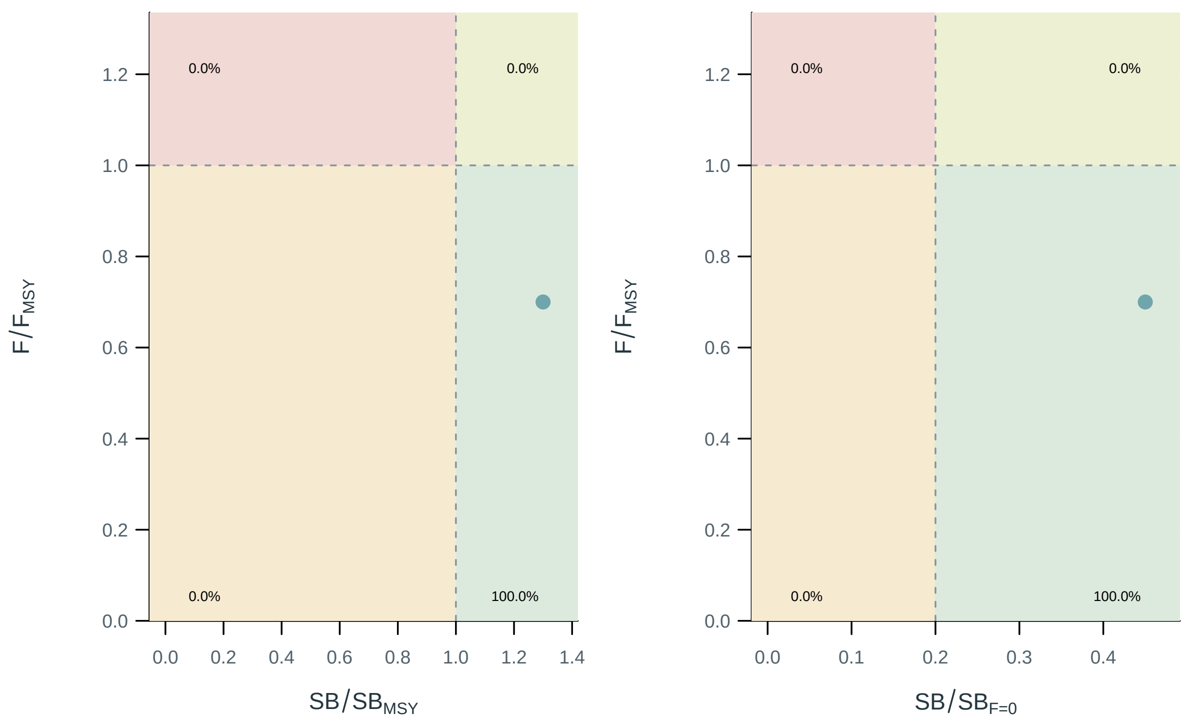

Figure 48: Stock status relative to the supplied spawning and fishing reference points.

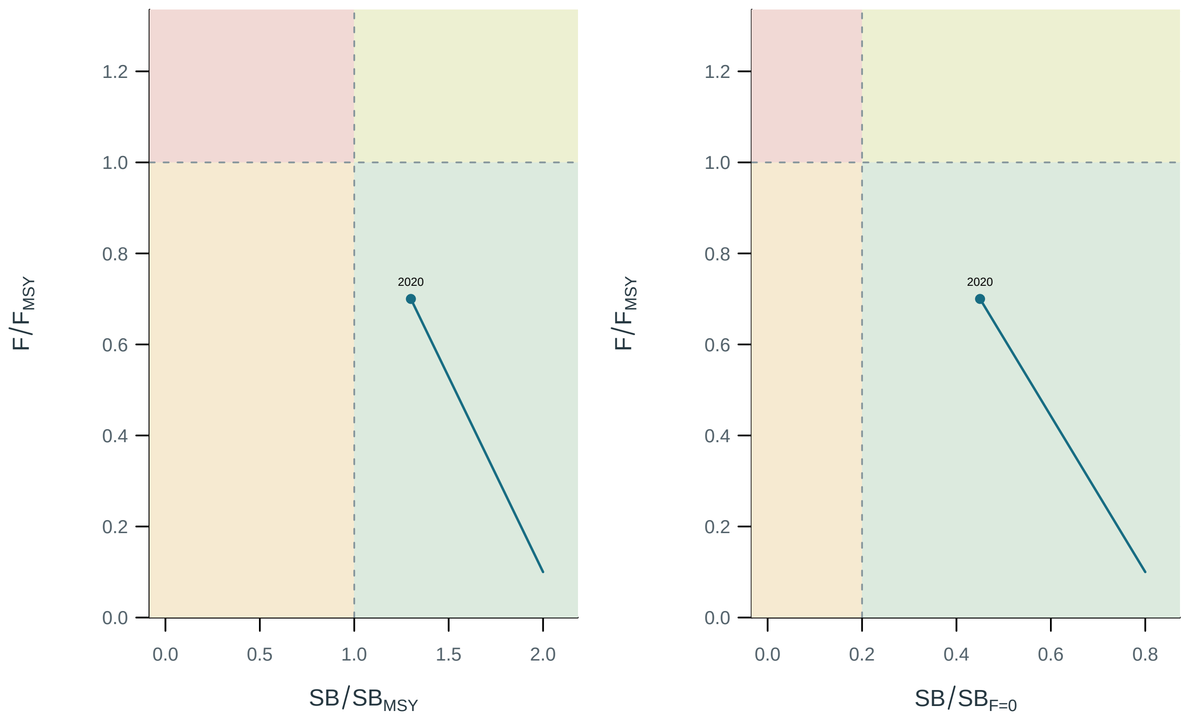

Figure 49: Stock status relative to the supplied spawning and fishing reference points.

Figure 50: Dynamic maximum sustainable yield and catch.

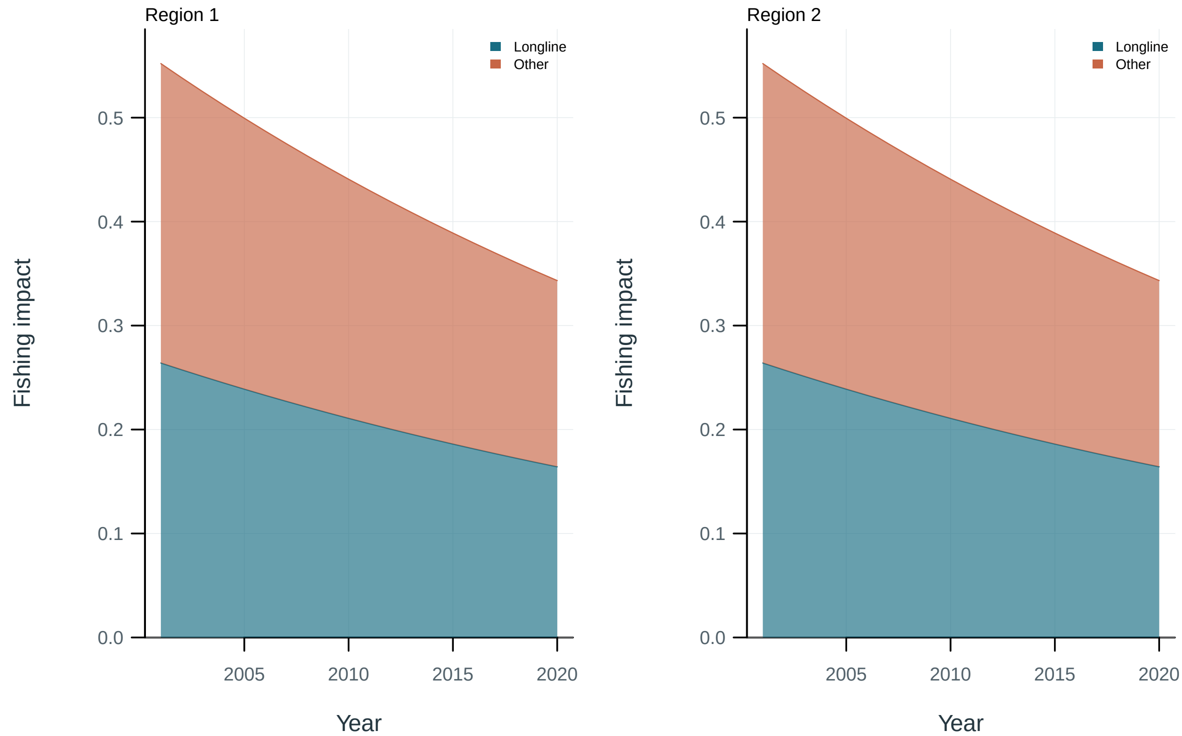

Figure 51: Illustrative output. Contributions of fishing groups to reduced spawning potential.

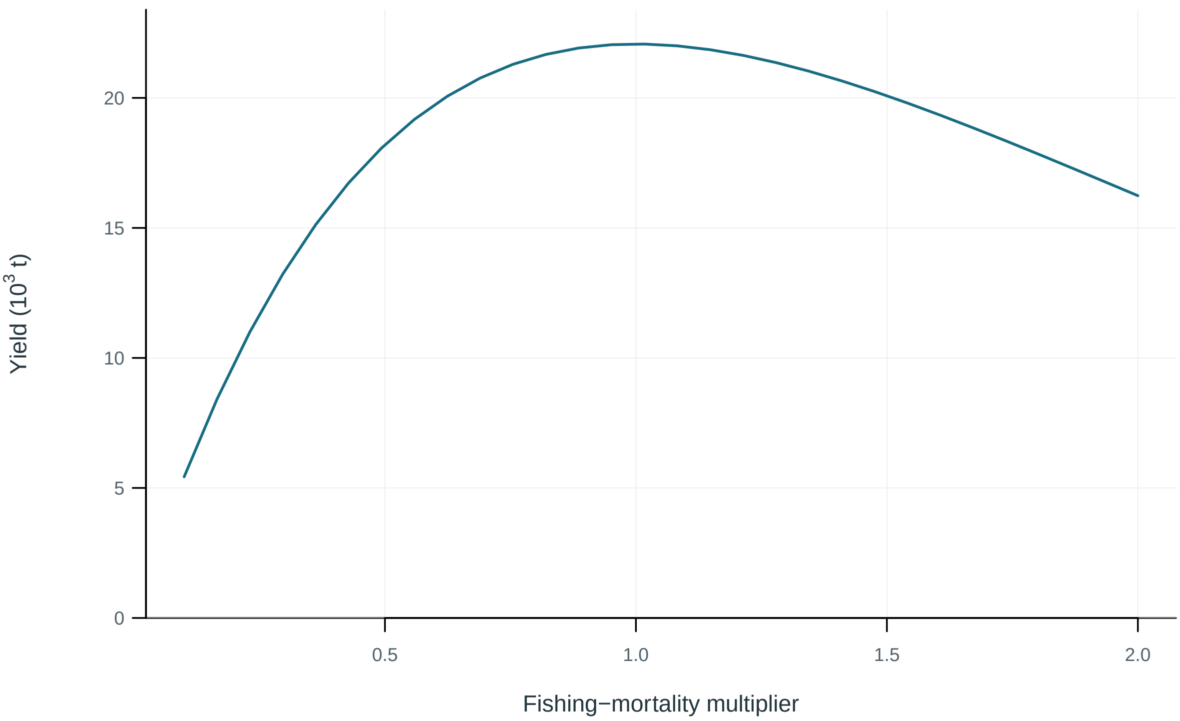

Figure 52: Equilibrium yield across fishing-mortality multipliers.

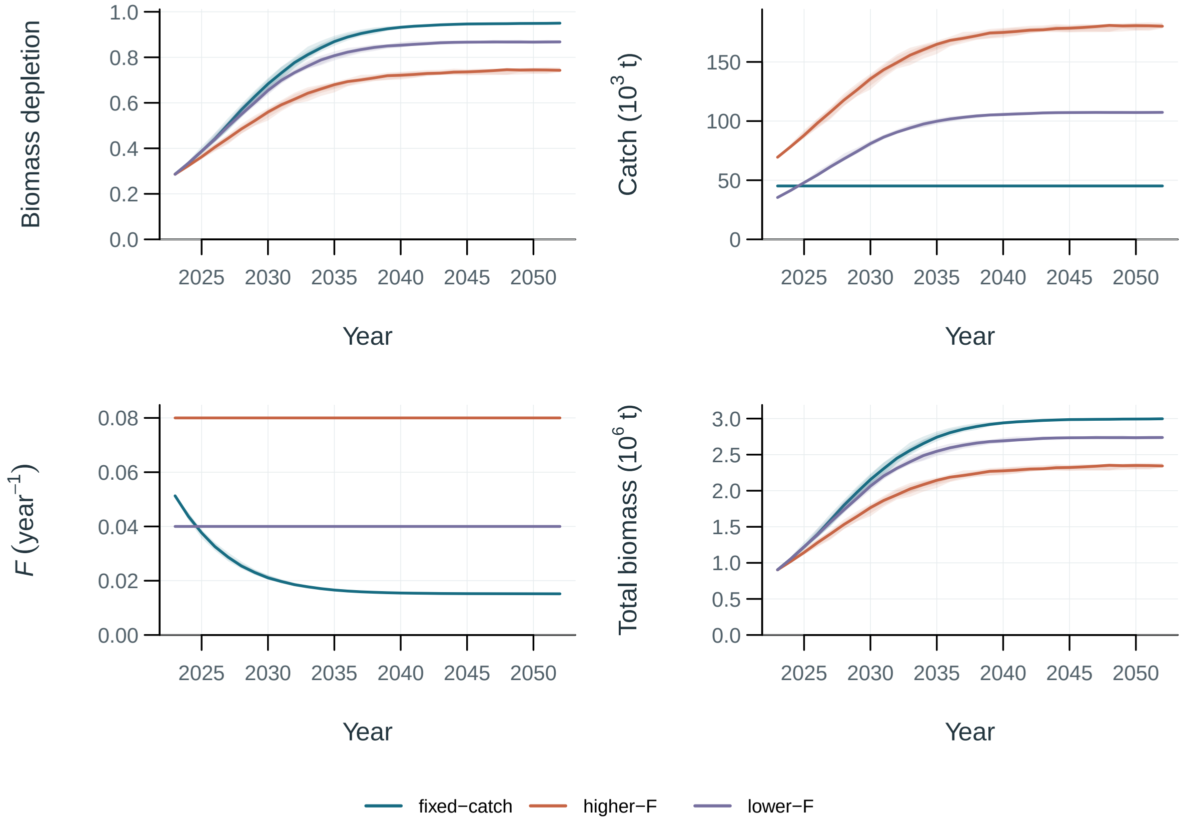

Figure 53: Illustrative surplus-production projections under fixed catch and two F scenarios. Lines are medians; bands span the central 50%, 80% and 95% of production-variability replicates.

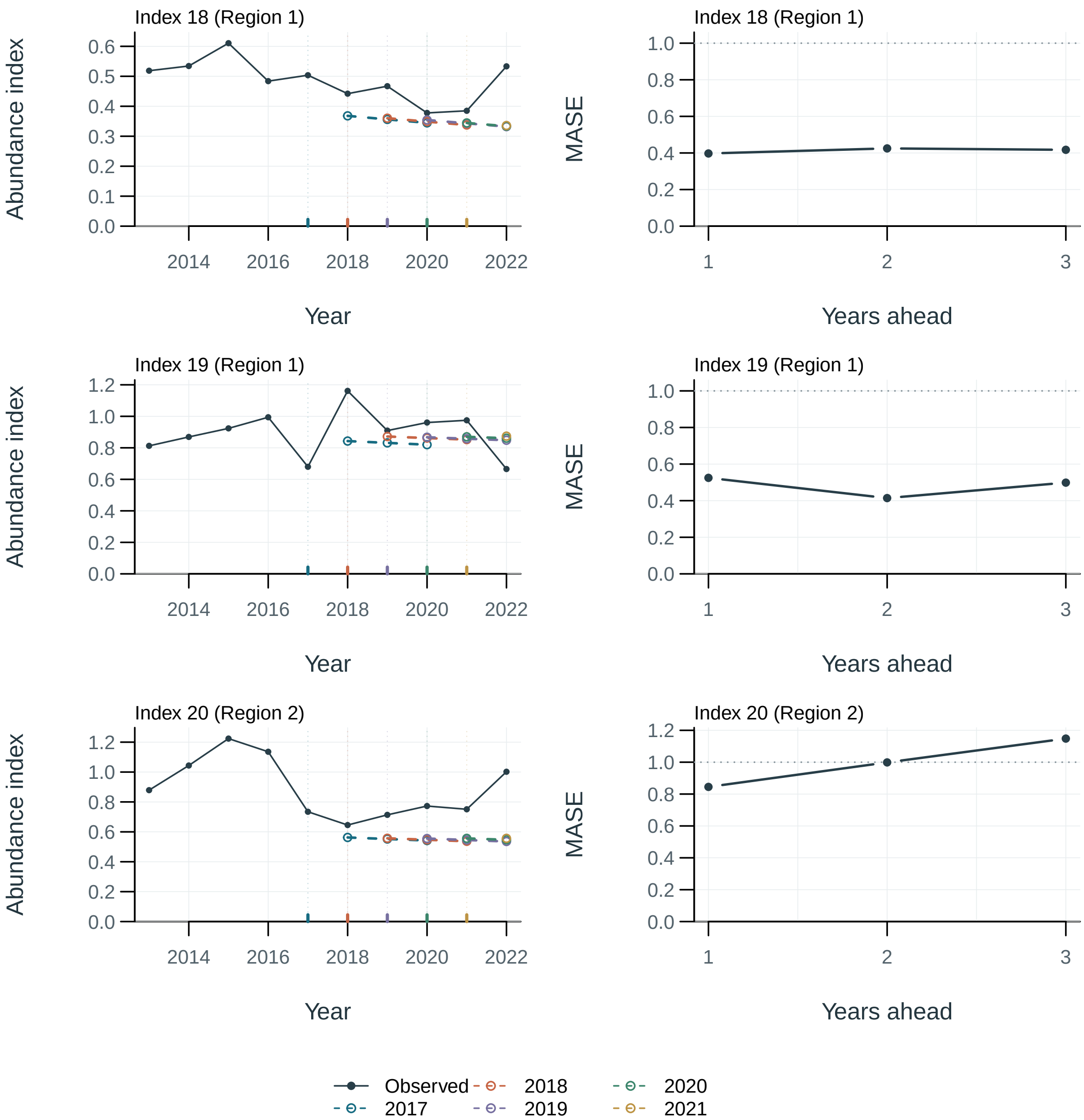

Figure 54: Observations (black) and held-out predictions (coloured dashed lines); vertical marks identify training cutoffs. Right panels show mean absolute scaled error (MASE) by forecast lead. The dotted line marks the training naive-error scale of one; lower values indicate smaller errors.

16 Supplementary outputs

ALB · Diagnostic model

ALB · Diagnostic model figures and tables

ALB · Uncertainty ensemble

ALB · Uncertainty ensemble figures and tables

Additional output formats · Display examples

Additional output formats · Display examples figures and tables

Hindcasting

Hindcasting figures and tables

Jitter analysis

Jitter analysis figures and tables

Likelihood profiles

Likelihood profiles figures and tables

Projection scenarios

Projection scenarios figures and tables

ALB · Retrospective analysis

ALB · Retrospective analysis figures and tables

ALB · Self-test recovery

ALB · Self-test recovery figures and tables

ALB · Sensitivities

ALB · Sensitivities figures and tables

Stepwise development

Stepwise development figures and tables

17 Appendices